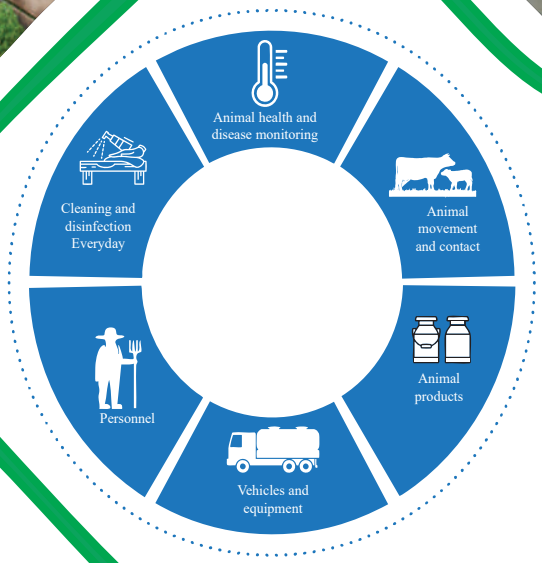
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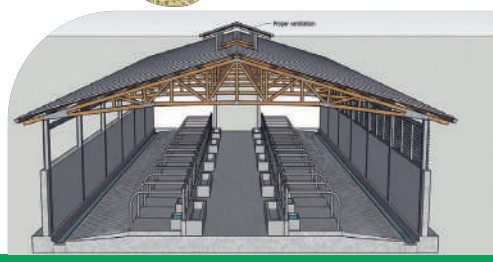
GOOD DAIRY HUSBANDRY PRACTICES AND BIOSECURITY MEASURES MANUAL











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GOOD DAIRY HUSBANDRY PRACTICES AND BIOSECURITY MEASURES MANUAL

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1. Livestock and Fishery Resource Development
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3. Animal Health and Veterinary Public Health
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Foreword

Ethiopia, home to the largest livestock population in Africa, finds itself at a crucial turning point in reshaping its dairy industry. As a fundamental pillar of the country's agricultural economy, the dairy sector plays a vital role in ensuring food security, sustaining rural communities, and driving economic growth. Under the Ethiopian National Dairy Development Strategy (NDDS) 2022–2031, the Ministry of Agriculture has embarked on an ambitious initiative: to multiply milk production fourfold from 7.1 billion litres in 2021 to 28.4 billion litres by 2031. Yet despite this vast potential, the sector continues to struggle with deep-rooted challenges that impede productivity, product quality, and sustainability. Issues such as limited good dairy husbandry practices and biosecurity measures in a country. These problems are exacerbated by the limited dissemination of technical knowledge and modern practices, particularly among the nation's smallholder dairy producers.

Recognizing these critical gaps, we emphasize that effective dairy production relies on optimized feeding strategies, enhanced animal comfort, successful genetic improvement, strict milk hygiene, and vigorous animal health. In light of this, we have developed this manual as part of our national initiative to address the sector's shortcomings and guide its transformation into a more efficient, resilient, and market-driven industry.

This guide offers practical, evidence-based recommendations for improved dairy husbandry, with particular emphasis on biosecurity measures. It is designed to equip dairy professionals, dairy producers and stakeholders with the tools and knowledge they need to boost productivity and ensure the safety and quality of dairy products throughout the value chain. Covering vital components such as advanced feeding and breeding strategies, improved housing, hygienic milk production and handling, and disease prevention, the manual bridges scientific research and on-the-ground practices.

We are confident that this manual will bring a noticeable improvement in the skills and attitudes of dairy producers, leading to higher production of hygienic milk to meet consumer needs and enabling them to earn better income. We offer our heartfelt thanks to the dedicated technical experts and development partners whose unwavering commitment and expertise have been instrumental in creating this manual, ensuring its relevance and quality.



Fikru Regassa /PhD/
State Minister,
Livestock & Fishery Resource Development Sector

Acknowledgement

This manual was developed through the collaborative effort of the Livestock and Fishery Sector State Ministry, whose leadership and guidance have been instrumental in shaping the content and ensuring its alignment with the national dairy development strategy.

We sincerely thank the Food System Resilience Program (FSRP) and the Livestock and Fishery Sector Development Project (LFSDP) for their financial support in enabling the efforts of the technical task force.
We also extend our gratitude to FSRP for its financial and technical support in facilitating the publication and contributions to enrich the manual with insights to promote good practices that support a sustainable, competitive, and resilient dairy sector.

We acknowledge the dedication and input of ministry technical experts, who contributed their expertise and practical experiences. Their active participation has ensured that this manual is both comprehensive and practical, meeting the real needs of dairy producers and extension workers.

Special appreciation goes to Ato Sagni Chemeda of the Ministry of Agriculture for his invaluable guidance and expert contributions throughout the manual development process.

Lead Executive of Livestock and Fisheries Development
Ministry of Agriculture

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Acronyms

AI	Artificial Insemination
AMR	Antimicrobial Resistant
BCS	Body Condition Score
BM	Biosecurity Measures
CES	Compulsory Ethiopian Standard
cfu/ml	Colony Forming Units per Millilitre
CIDR	Controlled Internal Drug Release
DM	Dry Matter
EIAR-HARC	Ethiopian Institute of Agricultural - Holeta Agricultural Research Center
FAO	Food and Agriculture Organization of the United Nations
GDHP	Good Dairy Husbandry Practices
ID	Animal Identity Card
ME	Metabolizable energy
TAI	Timed Artificial Insemination
TBC	Total Bacterial Count
TCC	Total Coliform Count
TDN	Total Digestible Nutrient

Definition of Terms

Biosecurity: A set of measures and practices designed to prevent the introduction and spread of diseases, pests, and harmful organisms among animals, humans, and the environment.

Biosecurity Measures of a Dairy Farm: - is the set of preventive practices designed to protect dairy animals from infectious diseases by controlling the movement of people, animals, equipment, and feed, and by maintaining proper hygiene and sanitation on the farm.

Commercial dairy farm housing: refers to the planned and well-structured buildings and facilities designed to provide a comfortable, clean, and safe environment for dairy cows in large-scale milk production systems.

Culling Practice of Dairy Farms: Culling in dairy farming refers to the process of removing cows from the herd, whether for sale or slaughter, and typically involves replacing them with new animals.

Dairy cluster housing: is a system where multiple dairy farms or producers are grouped in a specific area and share common facilities such as housing, milking, feeding, waste management, and biosecurity infrastructure.

Dairy Farm Record Keeping: The systematic process of documenting all activities, inputs, and outputs on a dairy farm, including animal health, milk production, breeding, feeding, and financial transactions, to improve farm management, monitor performance, and support decision-making

Dairy House Sanitation: Refers to the practices of keeping dairy animal housing clean, hygienic, and safe to prevent the spread of diseases, ensure animal health, and maintain the quality of milk and other dairy products. It includes regular cleaning, waste management, ventilation, and proper handling of bedding, feed, and water.

Dairy Sanitation Measures: Practices and procedures aimed at maintaining cleanliness and hygiene in dairy farms, including the handling of animals, milk, equipment, and the environment, to prevent contamination, ensure milk quality, and protect animal and human health.

Disease Control in Dairy Farms: The systematic practices and measures implemented on a dairy farm to prevent, manage, and reduce the spread of infectious and non-infectious diseases among livestock, ensuring animal health, productivity, and food safety.

Disease Prevention in dairy farm: The practices of implementing measures and strategies to reduce the risk of illness, control the spread of infections, and protect the health of individuals, animals, or populations before diseases occur.

Good dairy husbandry practices: A set of management and care practices applied on dairy farms to ensure the health, welfare, and productivity of dairy animals, while maintaining milk quality, environmental sustainability, and farm biosecurity.

Hygienic milk production: A practice of producing, handling, and storing milk under clean and sanitary conditions to ensure it is safe, high-quality, and free from harmful microorganisms, contaminants, and spoilage.

Isolation of the dairy herd: The practice of separating new, sick, or potentially infected animals from the main herd for a specific period to prevent the spread of diseases and protect herd health

Manure Management of Dairy: The systematic collection, storage, treatment, and utilization of manure produced by dairy animals to minimize environmental pollution, reduce disease risks, and convert waste into valuable resources such as fertilizer and biogas.

Smallholder Dairy Housing System: A smallholder dairy housing system refers to the design and construction of shelters for dairy animals managed by small-scale farmers, focusing on animal comfort, health, hygiene, and productivity while using locally available materials and simple technologies.

Zoonotic diseases: Diseases that are naturally transmitted between animals and humans and vice versa, either directly (through contact with infected animals) or indirectly (through vectors, food, or the environment).

1. Introduction

The dairy sector plays a vital role in supporting rural livelihoods, enhancing food security, and driving national economic growth in Ethiopia. However, its productivity and quality remain low due to persistent challenges in feed availability, poor breeding practices, inadequate housing and waste management, unhygienic milk handling, and limited adoption of health and biosecurity measures.

Feed scarcity, poor-quality forage, and traditional feeding practices heavily dependent on crop residues and natural pastures are major bottlenecks. Most smallholders lack access to year-round, nutritious feed, affecting both milk yield and animal health. Reproductive inefficiencies, caused by unplanned natural mating and limited use of artificial insemination, hinder genetic progress. Improving herd productivity requires well-defined breeding objectives, strengthened AI services, and targeted capacity building. Additionally, the housing practices range widely, but many smallholders lack appropriate structures for ventilation, hygiene, comfort, and waste disposal. Poor shelter compromises animal welfare, increases disease risk, and reduces productivity.

Health and biosecurity practices are also weak, leaving dairy herds vulnerable to infectious diseases. The absence of structured biosecurity plans and limited disease monitoring calls for the development of practical, locally adapted protocols and preventive health strategies.

Hygienic milk production and handling are often overlooked, with common issues such as dirty barns, unclean udders, and unsafe water use. This contributes to poor milk quality and public health risks. And the last is dairy record keeping faces several challenges, including low awareness and limited technical capacity among farmers, lack of standardized recording tools, poor access to digital technologies, and weak extension services.

This manual provides actionable solutions to these challenges, offering evidence-based guidance on improved feeding, breeding, housing, hygiene, health, and biosecurity management and record-keeping. By adopting these practices, Ethiopia's dairy sector can unlock its full potential, ensuring a more productive, resilient, and safe industry for the future.

1.1. Objective

To enhance dairy productivity through knowledge and skill transfer to experts, dairy producers, and other value chain actors on good dairy husbandry practices and biosecurity measures, thereby contributing to the sustainable food and nutritional security of the country.

1.2. Purpose of the manual

The purpose of this manual is to provide a comprehensive guide designed to strengthen dairy service delivery by equipping dairy producers, development workers, stakeholders, and service providers with the knowledge and skills needed to implement good dairy husbandry practices and biosecurity measures. The main purpose of this manual is to mainstream this practice across all scales of dairy farming, and it emphasizes six sections:- (1)Feed and feeding Practices ensuring dairy cattle receive balanced and adequate ration, (2) Breeding and Reproduction of dairy cattle are improving genetic potential and reproductive efficiency, (3) Housing and proper facilities to promote animal health and welfare, productivity, and operational efficiency, (4) apply proper Biosecurity measures preventing and controlling diseases in dairy cattle herds, (5) Hygienic Milk production and handling maintaining milk quality and safety and (6) record keeping to track animal performance, health, and reproduction, and to supports better management decisions. By engaging all relevant stakeholders through training and capacity-building, this manual supports the adoption of best dairy husbandry practices and the strengthening of biosecurity measures across the entire dairy value chain.

2. Feed and Feeding Practices of Dairy Cattle

Dairy cattle feeding practices are mostly traditional and depend on natural pasture, crop residues, and agro-industrial by-products. Dairy farmers face seasonal feed shortages, limited access to commercial feeds, and poor feed quality, which negatively impact milk production. To address these issues, integrating improved forage crops, adopting feed conservation, and promoting better feed management practices are recommended as sustainable solutions to enhance year-round feed availability and dairy productivity.

2.1. Nutrition Requirements of Dairy Cattle

Dairy cattle, particularly crossbred breeds, have specific nutritional needs that are often not met by common basal feed resources, which leads to significant production gaps. Addressing these deficiencies through targeted interventions is critical for normal body functioning, disease prevention, growth, weight gain, reproduction, and milk production. These requirements include a balanced intake of energy, protein, vitamins, minerals and water.

a) Energy Requirements

- Dairy cattle need energy for maintenance, milk production, growth, reproduction, health, etc.
- Dairy cattle should be supplemented with high-energy source feeds in addition to the basal diet of poor and medium energy feeds to meet their nutritional requirements for maintenance and milk yield
- Poor and medium energy feeds, like natural pastures, grass (Napier grass), mature hay, crop residues, green forages, legume straws, enset by-products, and sugarcane tops, comprise the basal diet. However, they are insufficient for optimal dairy performance due to low metabolizable energy (ME) and digestibility.



Figure 1. Medium and poor energy source feed stuff

- Dairy cattle should be supplemented with high-energy feeds like grains (maize, wheat, barley, sorghum), agro-industrial by-products (wheat bran, rice bran, brewer's grains, molasses), and roots (cassava, sweet potatoes) to meet their energy needs.



Figure 2. Good energy sources feed stuffs

Key points on feeding practices with grains and Agro-industrial by-products:

- Too little energy in the diet: Leads to weight loss, weakness, lower milk yield, increased illness risk post-calving, and smaller calves.
- Too much energy in the diet: Causes cows to gain too much weight before calving, raising the risk of difficult births, retained placenta, displaced abomasum, milk fever, ketosis, and mortality

b) Protein Requirements

- The protein requirements of dairy cattle depend on their age, weight, physiological stage (growth, pregnancy, or lactation), and production level.
- Improved dairy cattle should be fed high-protein sources such as legume forages (e.g., alfalfa, vetch, desmodium, lablab), agro-industrial by-products (e.g., Noug seed cake, cottonseed seed cake, linseed cake, brewery spent grain), and formulated concentrates to meet their protein needs.



Figure 3. Good protein source feed stuff

- Poor quality feed staffs like crop residues (teff straw, barley straw, wheat straw, maize stover, sorghum stover and finger millet straw), natural pasture (overgrown, fibrous, and poorly stored grasses), sugarcane tops, banana leaves, weeds and roadside grasses (usually unpalatable, contaminated, and low in nutrients) not meet protein requirements of dairy cattle and they should be avoided.

Key notes in protein-dense ration supplement practice:

- Too little protein in the ration: causes rapid milk production decline, excessive weight loss in milking cows, slower growth in calves and heifers, and smaller calves at birth.
- Too much protein in the ration: Excess protein in the ration is costly and wasteful, as surplus protein is broken down in the rumen and excreted.

c) Mineral Requirements

- Minerals are essential for growth, reproduction, milk production, immune function, and bone development in dairy cattle.
- Deficiencies can cause reduced milk yield, poor fertility and growth, lameness, and metabolic disorders (e.g., milk fever, retained placenta).
- Mineral requirements are higher in young growing animals, pregnant and lactating cows, and it's more important in early lactation
- Some minerals are stored in the body, e.g., iron in the liver and calcium in bones, while others are not, e.g., sodium, potassium, and have to be supplied in the feed ration.
- Minerals required by dairy cattle
  - Macro-minerals are needed in grams per kilogram of dry matter (g/kg DM) or % DM. They include calcium, phosphorus, magnesium, potassium, sodium, sulfur, and chlorine.
  - Micro-minerals are needed in milligram quantities (mg/kg DM or parts per million ppm) and include cobalt, copper, iron, iodine, manganese, zinc, selenium, and molybdenum.
- Farmers should adopt one of the following practices to meet dairy cattle's mineral requirements based on availability:Use locally available mineral premixes (like dairy licks or powders) in concentrates or feed troughs.
  - Offer salt licks or multi-nutrient blocks during late pregnancy and peak lactation, especially for smallholders.
  - Mix minerals into home rations (e.g., wheat bran, oilseed cake, mineral mix), adjusting based on milk yield and the cow's stage.
  - Feed mineral-rich forages and legumes (e.g., alfalfa, desmodium, lablab, cowpea, elephant grass, vetch) while avoiding low-mineral residues like straw.
  - Supplement calcium and phosphorus during late pregnancy and early lactation to prevent issues like milk fever.



Figure 4. Important sources of macro minerals

- Generally, grasses, legumes, silages, ground limestone, magnesium oxide, potassium chloride, and salt are key mineral sources of calcium, magnesium, phosphorus, sulfur, and trace elements.
- Home-mixed ration should contain calcium and phosphorus, which are the two essential minerals, and the optimum Ca: P ratio is 1.5:1 to 2:1.
- The reproductive and production effects of macro and micro mineral deficiencies in dairy cattle are described in Annex 1 in table 1.

d) Vitamin requirements

- Vitamins play a critical role in maintaining the health, productivity, fertility, and immunity of dairy cattle at different stages of production
- Dairy cattle at different production stages should receive vitamin-rich feeds, including green forages (Napier, desho, lucerne), legume forages (vetch, cowpea, alfalfa, lablab), root crops (sweet potato vines, cassava leaves), and agro-industrial by-products (wheat bran, brewer's grains, molasses).
- If green forage or vitamin-rich feed is unavailable, especially during the dry season, use commercial premixes or mineral-vitamin blocks to meet vitamin needs. Essential vitamins for dairy cattle, their sources, and deficiency consequences are detailed in Annex 1 table 2.

e) Water requirements

The body of a dairy cow is composed of 70–75% water. Milk is about 87% water. Water is not a feed as such because it does not provide specific nutrients. However, it is essential in body processes and to regulate body temperature. Water is involved in digestion, nutrient transfer, metabolism and waste removal. Water has structural and functional roles in all cells and all body fluids. An abundant, continuous, and clean source of drinking water is vital for dairy cows. The amount of water required per day depends on many factors, including feed (Dry matter, protein and mineral content & balanced ration), animal factors (breed, age, body weight, stage of production and health status), environment management practices (surrounding temperature, water temperature and season of year),(watering frequencies, watering utensils, & proper housing), water quantity and quality (organoleptic-physiochemical, microbial, and toxic substance) and milk production.



Figure 5. Factors affecting dairy cattle's water intake

2.2 . Feeding different categories of dairy cattle

Dairy cattle feeding practices depend on their physiological category (calves, heifers, dry cows, lactating cows). Proper nutrition is essential for optimizing productivity and maintaining animal health.

2.2.1. Feeding Management of Lactating Cows

Optimal feeding management of lactating cows not only affects milk yield but also plays a critical role in maintaining animal health, ensuring reproductive success, and economic returns on the farm. The key target of feeding lactating cows is to:

- Maximize milk production by meeting their nutritional requirements, which should be balanced according to the stage of lactation, milk yield, pregnancy status, and maintenance needs.
- Ensure adequate preparation during the dry period to support fetal development, restore body energy reserves, and promote the regeneration of mammary tissue in readiness for the next lactation cycle.

Understanding the Lactation phases: Cows must calve to produce milk, and the lactation cycle is the period between one calving and the next. The lactation cycle is typically divided into early, mid, and late lactation, each with distinct nutritional needs. A number of changes occur in cows as they progress through different stages of lactation. As well as variations in milk production, there are changes in feed intake and body condition, and stage of pregnancy.

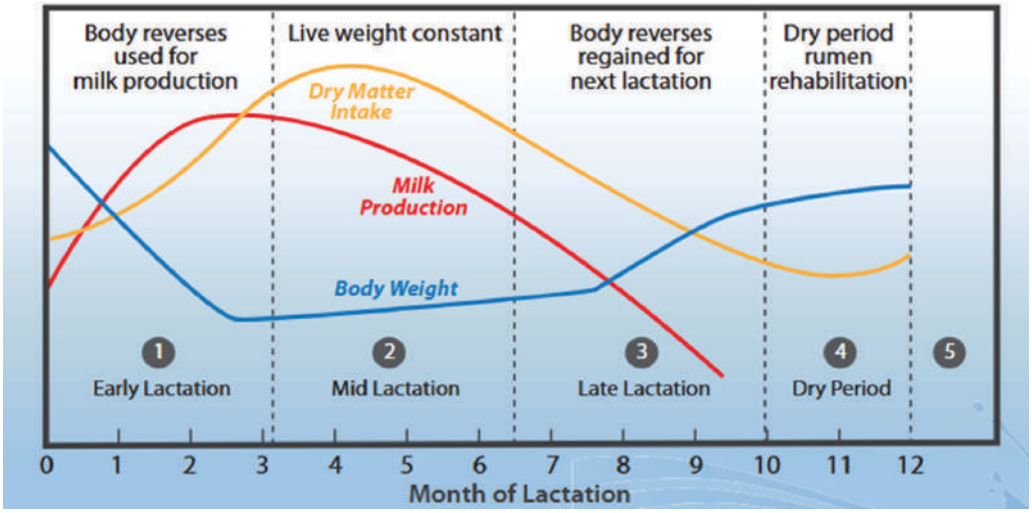


Figure 6. Lactation phases (Adapted from Moran, 2013)

i. Feeding practices in early lactation (1-70 days)

- The critical importance of proper feeding in early lactation is to achieve and sustain peak milk yield.
- During this phase of lactation, milk production increases rapidly so that the voluntary feed intake cannot meet the energy demand. This results in an energy deficit leading to use of body reserves and to weight loss (negative energy).
- At this stage of production, cows should receive high-energy, high-protein feedstuffs like quality forage and hay ad libitum or at 3–3.5% of their body weight in DM/day to support milk production, health, and reproduction. Additionally, provide supplementary concentrate based on the cow's milk yield, as shown in table 2. For example, for A cow body weight of 450 kg and producing 20 litres of milk per day daily feed requirements are calculated in the following table for easy practice.

Table 1: Example of feeding practice for early lactating cow

Parameters	Value
Body weight	450 kg
Milk yield/day	20 litters
DMI of the cow/ day	3–3.5% of her body weight, 13.5-15.7 kg DM/day
As-fed Fresh forage (assume that 20 % DM	67.5 - 78.8 kg/ day
As-fed hay (assume that 80% DM)	16.9 - 19.7 kg/ day
Silage (assume that 30% DM)	45 - 52.5 kg/day
Concentrate supplement to support 20 litres of milk yield/day. For a cow producing 13 to 28 liters of per/day, it is recommended to provide 1kg of concentrate supplements for every 2.5 liters of milk produced.	8 kg concentrate per day,

- Whenever possible, use hay from improved forages like desho grass, Rhodes grass, Panicum, alfalfa, oats, vetches, cowpea, lablab, and tree lucerne
- For lactating cows, fresh forage or silage should be made from high-quality cultivated forages suitable for cut-and-carry or silage, such as Guatemala grass, elephant grass, maize, sorghum, sweet potato vines, tree lucerne, lablab, oats, vetches, desmodium, cowpea, desho grass, Rhodes grass, and alfalfa.
- A grass-to-legume ratio (70:30) in silage optimizes fermentation quality and nutrition. Grass supplies sugars for fermentation, while legumes contribute protein, enhancing preservation, animal intake, and reducing spoilage risks compared to high-legume mixtures. Hence, the recommended feedstuffs and feeding practices for early lactating cows, applicable from smallholder to large-scale commercial dairy farming systems indicated in Annex 1 table 3.



Figure 7. Lactating cows are fed on TMR and corn silage

- Precaution: Using over 60% DM from concentrates can cause ruminal acidosis. Forage should be the main component of a cow's feed ration; it is cheaper than concentrates and boosts farm profit.

ii. Feeding practice in mid-lactation (71–150 Days).

- The critical importance of proper feeding practice in this phase is to maintain peak production efficiency.
- A cow with a body weight of 400kg and producing 13 litres of milk daily requires about 12 kg of dry matter (DM) per day, which is 3% of her body weight. This can be achieved with approximately 60 kg of fresh forage (20% DM), 15 kg of hay (80% DM), or 40 kg of silage (30% DM). Additionally, a concentrate supplement of 5.2 kg per day is needed to support the milk production described in table 2.
- At this production stage, high-quality feed, forages, silage, or hay should be prepared from improved feeds under early lactation.
- In case of feed scarcity, use natural pasture hay, treated crop residue, or residue mixed with protein byproducts, and provide supplementary concentrate.
- Farmers should be well aware of ensuring adequate fiber intake to maintain rumen health.

iii. Feeding practice in late lactation (151–305 Days)

- The critical importance of proper feeding in this stage of lactation is to replenish body reserves and support pregnancy.
- It is recommended to provide forage, hay, or silage similar to that used during mid-lactation. If these are limited, alternative feed resources like natural pasture hay, treated crop residues, and conventional feeds such as banana pseudo-stem and leaves should be included in the dairy feed ration. Therefore, the recommended feedstuffs and feeding practices for late lactation are applicable across smallholder and large-scale commercial dairy farming systems as indicated in Annex table 3. In these inferences, recommended supplementary feeding guidelines for lactating cows, with concentrate supplementation levels corresponding to milk yield described in the following table 2.

Table 2: Milk yield corresponding to the concentrate supplementation amount

Milk Yield (L/day)	Concentrate Allowance (kg milk/kg concentrate)
<13	3:1
13–28	2.5:1
>28	2:1

Noted: Up to 7 kg of milk/day from forage. Add 1 kg dairy meal per 1.5 kg extra milk

Water Requirements for Lactating Cows: Water is essential for milk production, nutrient transport, waste excretion, and temperature regulation. Even slight restrictions cause rapid declines in feed intake, milk yield, and immunity. Therefore, lactating cows should have 24-hour access to clean water. Precision Calculation Formula: (Body Weight in kg / 10) + (1.5 × Daily Milk Yield in litres) = Total Daily Water (litres). However, this can increase to as high as 5 litres for every litre of milk in hot conditions. Example Calculation for a 325 kg cow producing 10 litres/day: (325 ÷ 10) + (1.5 × 10) = 32.5 + 15 = 47.5 litres/day

2.2.2. Feeding practice during dry period

Proper feeding during this period is crucial for rebuilding mammary tissue and body reserves, ensuring quality colostrum for the calf, supporting calf development, and preparing the cow for increased milk production while preventing issues like milk fever and ketosis.

- It's about 60 days before the next calving
- During the 8th Month of Pregnancy: the cow should be maintained on good pasture or supplement poor pasture with 1–2 kg/day of concentrate
- During the 9th Month of Pregnancy: the cow could be maintained on good quality roughage and supplemented with 1–2 kg of concentrate per day. In case of poor-quality roughage, the cow could be fed supplementary concentrate 3–4 kg/day.
- Control calcium intake in the last 2 weeks pre-calving
- Key notes in dry cow management: Focus on BCS management (Target 3–3.5); adjust concentrate feeding based on pregnancy and roughage quality; offer roughage ad libitum; implement strategic mineral withdrawal; ensure consistent water access.

2.2.3. Feeding management of calves and heifers

A. Feeding management of calves

The critical importance of proper feeding pre-weaning calves is to ensure robust health, meet all nutrient requirements, significantly reduce mortality rates, and maintain a minimum growth rate of 400 grams per day.

a. Colostrum feeding

- Colostrum feeding is essential to provide passive immunity to newborn calves.
- Colostrum should be given as soon as possible after calving, within 30 minutes to 2 hours
- The calf should receive 8–10% of birth weight per day, divided into multiple feeds, for the first seven days of life.
- The calf ability to absorb antibodies declines rapidly, dropping by approximately 50% after just 6 hours. Delayed feeding severely compromises the calf's immune protection.
- If the bottle-feeding method is applied, the temperature should be maintained between 39–40°C, as well as fed immediately after milking or gently warmed if stored.
- Overfeeding of colostrum can lead to diarrhea. Therefore, precautions should be taken in the feeding management of a newly born calf.
- Artificial colostrum: Where maternal colostrum is unavailable, an emergency substitute can be made using: 1 egg + 400 ml warm water + 600 ml whole milk + 1 teaspoon castor oil. The calf should be offered this mixture three times daily for the first four days.
- Feeding equipment (bottles, nipples, buckets, etc.) should be cleaned and sanitized after every use to prevent disease transmission
- Sick calves should be fed last to minimize the risk of spreading pathogens to healthy animals via equipment or the handler.



Figure 8. The newborn calf is going to suckle colostrum from the dam

b. Milk Feeding:

- Calves should initially receive whole milk at a rate equivalent to 10% of their body weight daily. This provides essential nutrients and energy for early growth.
- Solid feeds are gradually introduced into the diet (typically starting at 2–3 weeks), and milk volumes should be systematically reduced to encourage intake of other feeds and support rumen development.

c. Milk Replacer Feeding

- It is an alternative solution when whole milk is scarce or cost-prohibitive; nutritionally balanced milk replacers offer a viable alternative.
- It can be stored long-term as powder and mixed with water just before feeding.
- Make sure that during procurement of the milk replacers crude protein content is 18-20%
- Provided the calf milk replacer is formulated correctly from good-quality ingredients and fed according to the instructions, which are usually on the calf milk replacer bag, calves can grow equally well when reared on calf milk replacer, and their rumens can develop just as well as they would be on a diet of whole milk.

d. Roughage and Calf Starter Feeding:

- Newborn calves should be fed good quality hay/green (mixture of oat and vetch hay, alfalfa, cultivated/native hay, napier grass, etc. from the first week onwards.

- The amount of hay to be fed to calves depends on the amount of concentrate required to be provided.
- Calves should not be provided with dusty, extremely fibrous, and mouldy hay.
- The calf starter ration should be kept in a hygienic environment.
- Calf starter ration should provide the required amount of energy, protein, minerals, and vitamins. In this regard, calf starter ration can be sourced from feed processors/feed millers or homemade using the following ingredients, and the composition is indicated in table 3.
- Precaution: Avoid grazing calves on parasite-contaminated pastures until they develop stronger immunity.

Table 3. Sample composition of calf starter ration

Ingredient	Percentage (%)
Wheat bran	40
Wheat middling	15
Noug cake	40
Lime stone	4
Salt	1

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e. Water requirements of calves

Water catalyzes rumen development; calves under three weeks need small, frequent clean water servings daily, while after three weeks, they require at least 4 liters per kilogram of dry starter feed.

f. Calves Feeding Schedules and Weaning

Pre-Weaning Feeding Strategy: In this phase, solid feed should be increased gradually to stimulate rumen development. Offer 0.5–1 kg/day of high-quality starter feed alongside ad libitum access high quality roughage. This approach ensures balanced nutrition and prepares the calf's digestive system for eventual weaning. Therefore, the feeding schedule options for calves for intermediate feeding and early weaning feeding are described in Annex 1 tables 4 and 5.

Table 4: Sample of a feeding schedule for calves

Age of calf (days)	Milk kg/day	Total Milk (kg)	Calf starter (kg/d	Roughage
1 to 7	4(Colostrum)			
8 to 21	5	70	Handful	
22 to 42	6	126	0.5	Yes (ad lib)
43 to 56	5	70	0.5	Yes (ad lib)
57 to 63	4	28	1	Yes (ad lib)
64 to 77	3	42	1	Yes (ad lib)
78 to 84	2	14	1.5	Yes (ad lib)
Wean the calf				
Total		**350**	**55**	

- **Calf weaning:** Calf weaning is the gradual process of transitioning calves from milk and milk replacer to become fully dependent on other feeds. Calf should be weaned when the calf has reached twice its birth weight and consistently consumes dry starter feed at or above 1.5% of its body weight

B. Feeding management from weaning to breeding age

- After weaning, heifer growth continues with high rates of muscle weight gain (protein) and low rates of adipose tissue.
- The critical importance of proper feeding in this stage is to:
  - Reduce the interval between weaning and first lactation to boost genetic progress.
  - Maintain growth rates of 500 -700 grams/day and minimize mortality with balanced nutrition.
- Reach at least 230 kg by 14–16 months for successful breeding, and the heifer's body weight at breeding should be 55-60% of a mature body weight
- Reach early calving at 22–24 months to maximize milk yield and herd economics.
- Mammary gland development is faster between three and nine months of age and is highly influenced by nutrition during this period.
- Pre-breeding heifer ration should be formulated from high-protein source feed stuff to support optimal lean tissue growth.
- Excessive energy intake prior to breeding should be avoided to minimize unwanted fat tissue accumulation.
- Heifer body condition should be monitored closely for signs of excess fat deposition, especially when targeting rapid weight gain for earlier breeding and calving.
- Feeding options based on a concentrate-to-roughage ratio of 50:50 can be used to support the optimum growth of cross-bred dairy heifers.
- For confined/zero-grazing systems, prioritize ad-libitum roughage access
- Provide concentrate at 1% of body weight (BW) or adjust to 1–4 kg/day based on age, size, and Roughage quality



Figure 9. Weaned calf supplemented with formulated feed

C. Feeding management of heifers from breeding to calving

i. Post-breeding 1 -5 months

- The heifer should be offered high-protein and energy source feed stuff to ensure maintenance and growth
- Recommended best practice: from post-breeding to mid-gestation, the heifer should be offered at least 2.0–2.5% of body weight in DM/day.
- Offer good-quality roughage basal feed (crop residues like teff straw, maize stover) supplemented with urea-molasses or legume forages (e.g., vetch, lablab, alfalfa) and Cottonseed cake, noug seed cake, or brewery by-products as a protein source
- Offer mineral licks or premix with essential trace elements (Zn, Cu, Co, Se, I).

ii. The last 3 to 4 months before calving

- Last 2 -3 months before calving, fetal growth accelerates; nutritional demands rise sharply
- Pregnant heifers over six months should be offered improved forages like Napier, desho, and elephant grass intercropped with legumes, plus 1–1.5 kg/day concentrate supplementation.
- The amount of concentrate proportion should be increased to support both the dam and fetus.
- The body condition score should be maintained between (BCS 3.0–3.5), and over-conditioning should be avoided to reduce calving and metabolic problems.
- Ensure adequate calcium, phosphorus, and trace elements (Se, I, Zn) and supplement with salt licks or commercial mineral mix

iii. Pre-calving (last 2-3 weeks)

- The lactation ration should be gradually introduced to prepare the rumen and metabolism for milk production.
- Reduce bulky feeds to make room for the fetus while increasing nutrient density.
- Offer small but frequent meals of good-quality forage/hay and concentrate.
- Water: Provide clean water freely, especially in hot, lowland areas.

Water Requirements of Heifer: Crossbred heifers require increasing amounts of water as they grow: 10–15 litres daily for calves (3–6 months), 15–20 litres for yearlings (6–12 months), and 20–25 litres for heifers aged 1–2 years. Key factors such as climate can increase water needs by 20–30%, while feed type also influences consumption, with high-concentrate diets or dry forage demanding more water. Therefore, water provision should be adjusted accordingly to meet these varying requirements.

2.3. Monitoring the feeding practices

- Concentrate feeds are rich in protein, energy, and minerals but low in fiber, offering limited ruminal stimulation, and they may lead to rumen acidosis if they exceed 60% of the diet's dry matter. In contrast, forage feeds like alfalfa and maize silage are fibre-rich, support optimal rumen function, provide essential nutrients, and are more cost-effective, making them ideal as the main component of a dairy cow's diet.
- To assess whether our feeding practices are appropriate or not, monitor cow signals (key parameters) and evaluate how well the ration is balanced.

i. Rumen fill:

- The rumen fill of a lactating cow serves as an important indicator of feed intake and digestive health.
- Poor quality forages/roughages will fill up the rumen without meeting the nutrient requirements of the cow producing a lot of milk. E.g., Poor quality hay and crop residue straw.
- The rumen may look full due to the bulkiness of fibrous, low-digestibility feed, but the cow remains undernourished.
- Rumen fill resulting from poor-quality roughage leads to several adverse effects, including limited additional feed intake due to the physical fill effect, inadequate nutrient absorption, reduced milk yield and poor lactation performance, loss of body condition, impaired rumen function, hard or dry manure due to low moisture and poor rumen fermentation, and weakened immune and reproductive functions.



Figure 10. Cows maintained on poor-quality feed

ii. Body Condition Score

- Cows with too low body condition scores (BCS) have not consumed enough energy in the past month, whereas cows with excessively high BCS have consumed too much energy during the same period. It is important to note that the ideal BCS varies with the stage of lactation.

iii. Manure

- Watery or thin manure may indicate an excess of concentrates, protein, or energy in the diet; in such cases, the amount of protein-rich concentrates should be reduced, and the fiber (forage) content should be increased.
- Very thick manure may indicate excessive fiber and insufficient protein or concentrate in the diet, so ensure the cow has adequate water intake, increase the proportion of protein-rich feed, such as high-quality forage, alfalfa, or concentrates, and reduce the amount of poor-quality roughage like straw or low-grade hay. Dehydration can also cause cows to produce very thick manure, making water availability essential.

iv. Manure digestibility

- Observe the manure to identify any undigested components; this reveals which parts of the feed the cow is unable to digest.
- If whole grains of maize, wheat, or oats are visible, try to crush them before feeding to improve digestibility.
- If large particles are present, chop the forage into smaller pieces and provide high-quality forage that is more easily digestible
- If numerous crushed concentrate particles are seen in slightly thin manure, it indicates rapid rumen passage. To slow this down, increase the inclusion of high-quality forage.

v. Feed quality variation

- Forages harvested at a later maturity stage contain much less protein and energy content.
- Some bran contains a lot of seed particles, while other bran has only the hulls (fibre).
- Some concentrate meals contain high proportions of oilseed cakes (protein), whereas others have small amounts.

2.4. Dairy cattle body condition scores (BCS)

Body condition scores are the visual evaluation of the amount of muscle and fat covering the bones of an animal. It can be used to assess the feeding regimes in dairy cattle.

- It can be assessed independently of live weight, gut fill, and pregnancy status and involves observing specific points on the animal.
- Scoring enables farmers to compare the condition of their cows with recommended targets.
- BCS is a visual and physical assessment (1–5 scale) of subcutaneous fat reserves, indicating energy balance. Importantly, the target BCS of cows varies depending on the stage of lactation.
- 1= Emaciated (bones visible) and
- 5= Obese (no bones palpable)

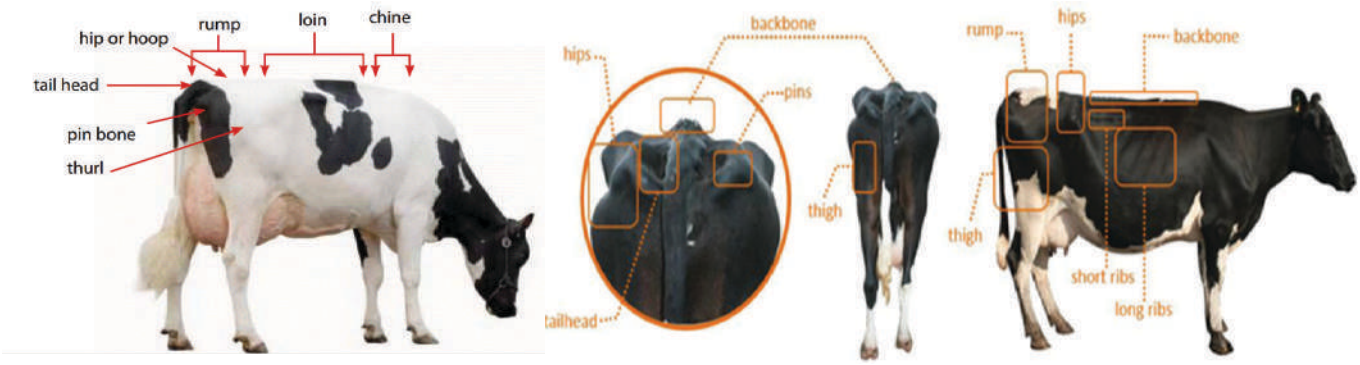


Figure 11. Critical points for body condition scoring

Table 5. Dairy Cow body condition scores

Score and Conditions	Description	Pictorial description
1,Emaciated	***Short ribs*** • Ends sharp to touch • Loin prominent, shelf-like appearance • Obvious scalloping over top and ends ***Backbone*** • Vertebrae prominent in chine, loin, and rump area • Individual bones are easily visible ***Hook and pin bones*** • Sharply defined, very angular in appearance • No discernable fat pad ***Tail head*** • Sunken and hollow on either side of tail head with obvious folds of skin • Ligaments connecting pin bones to spine are sharply defined • Vulva prominent	
2, Tin	***Short ribs*** • Ends not as prominent as BCS 1, but can be felt • Edges have slight fat cover, and slightly more rounded appearance • Overhanging shelf effect less apparent ***Backbone*** • Vertebrae in chine, loin and rump area, less visually distinct • Easily feel individual vertebrae ***Hook and pin bones*** • Bones still prominent, angular • No fat pad palpable ***Thurl (area over pelvis):*** • Less severe "V shaped" depression • Little tissue cover ***Tail head*** • Both sides of the tail head are sunken and hollow • Sharply defined ligaments connecting pin bones to spine	
3, Average	***Short ribs*** • Ends can be felt with moderate pressure • Ribs appear smooth without noticeable scalloping • Overhanging shelf effect much less apparent ***Backbone*** • Vertebrae in chine, loin and rump area appear rounded • Backbone visible, but individual vertebrae not distinct ***Hook and pin bones*** • Visible, but smooth, with rounded appearance ***Thurl (area over pelvis)*** • Forms "U shaped" depression ***Tail head*** • Both sides of tail head somewhat hollow, but skin folds not distinct • Ligaments connecting pin bones to spine are rounded in appearance	

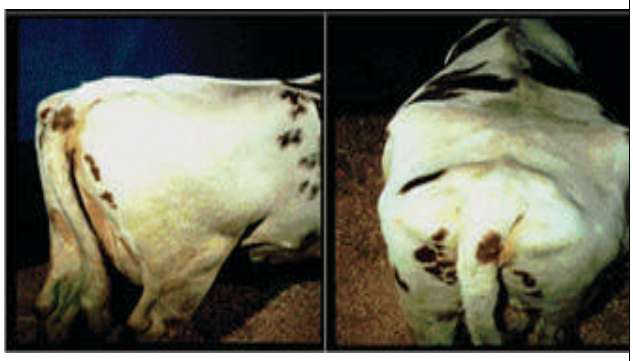
4, Heavy	*Short ribs* • Individual rib ends not visible • Overhanging shelf effect slight, barely visible *Backbone* • Vertebrae in chine rounded, smooth • Loin and rump areas appear flat *Hook and pin bones:* • Rounded, with obvious fat covering *Thurl (area over pelvis):* • Area between hooks and pins almost flat • Pelvic bone only felt with firm pressure *Tail head* • Sides of tail head not hollow, no skin folds	
5, Fat	*Short ribs* • Ends can't be seen • No overhanging shelf effect *Backbone* • Vertebrae in chine, loin and rump not visible • Difficult to feel individual vertebrae *Hook and pin bones* • Very round, buried (almost disappearing) in fat tissue *Thurl (area over pelvis):* • Appears flat • Filled in between the hooks and pins *Tail head* • Hollow filled in • Areas on both sides of tail head buried in fat tissue	

Table 6. Recommended BCS for different lactation stages of dairy cattle

Lactation stage	Recommended target BCS
Dry off	3.25-3.75
Calving.	3.25-3.75
Early lactation	2.50-3.25
Mid-lactation	2.75- 3.25
Late lactation	3.00 - 3.50
Growing heifers	2.75 - 3.25
Heifers at calving	3.25 - 3.75

Nutritional Management

- Adjusting dietary intake based on BCS is crucial.
- If a cow has a BCS < 3.0, increasing energy intake can help replenish energy reserves.
- If a cow has a BCS exceeding 3.75, energy intake should be reduced to avoid excessive fattening.
- Ideal scores fall in the range of 3.0 - 3.75 at dry off and calving, and 2.5 - 3.25 at peak lactation.
- Cows should not change by more than 1 condition score class over any lactation period

3. Dairy Cattle Breeding and Reproductive Practices

Ethiopian dairy farmers face significant difficulties in sourcing high-quality replacement heifers due to the limited availability & high cost of exotic breeds (e.g., Holstein/Jersey) & their crossbred stock in the local markets. Without reliable access to productive replacement heifers, smallholder farmers struggle to transition from low-yielding indigenous breeds to more productive exotic or crossbred animals, perpetuating low milk yields and hindering sector growth.

3.1. Selection Criteria of the dairy Cattle

Proper selection is the first and most important step to be adopted in dairying. Records are the basis of selection and hence proper identification of animals and records keeping are essential. The major parameters for the selection of dairy cow are as follows:

a) **Conformation (body shape):** The ideal dairy cow shows the ideal dairy character, shape and frame. Cow with good physical attributes like eye, udder development, feet and legs, and overall body condition. The following conditions must be met when selecting dairy cows based on their body shape:

- Legs are straight, proportional to her body and has correct hoof.
- The central or median suspensory ligament of the udder must be extremely strong and well attached.
- It has a wide deep and uniform udder, teats well placed, not too low below the hocks and attached high in the back and far forward underneath.
- The four teats are not covered and equal.
- Teat size (over and undersized teats should be avoided); shape and placement are highly heritable.
- Should have a posture of slimmer on the front while wider from behind(triangular)
- Should be alert and be clearly visible eyes.
- Its backline is straight and her rear end is wide and square which will facilitate easier calving and shows strength.
- Should have optimal BCS between 2.75 and 3.5 (on 1-5 scale) with no excess fat (>3.5 BCS)

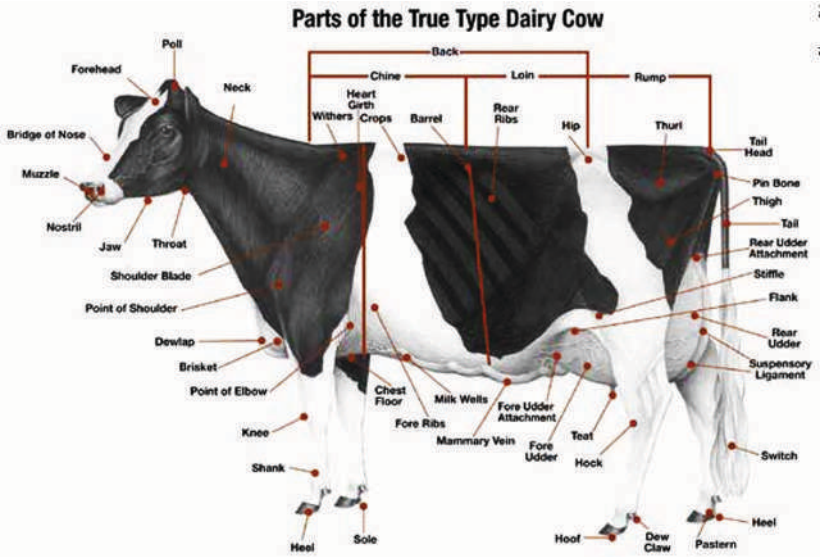


Figure 12. An ideal dairy cow for selection

b) Pedigree/family history: Pedigree/family history means that selecting dairy cattle based on the information obtained from dams, sires, sisters, brothers or any higher level of blood-related animals by analyzing the reproduction record.

- Traits we measured have high heritability selection done in this way is more reliable and proven.
- Traits we measured by pedigree selection are; Amount of milk production, milk fat content, calving rate, disease resistance and adaptability of the environment.

c) Select based on individual performance: Selecting dairy cow based on analyzing the previous reproduction & production records.

- Properly check for age and the parturition history of the individual animal.
- Cows showing milk production increment progress, produce a calf per year should be selected.

d)Health status: free from physical injuries, feet and hoof problems, any disability that can interfere with pregnancy and delivery. Select by visual observation and in consultation with health experts as well as analysis of health records.

e)Temperament/ Behavior: Avoid animals with bad behavior's such as head and leg kicking, and those that avoid their calf to suckle and difficult for milking.

f)Exotic Blood Level Dairy Cattle selection: Select pure exotic and improved breed dairy cattle based on husbandry experience, which is a critical determinant of farm productivity, sustainability, and profitability. While higher exotic genetics (e.g., Holstein, Jersey) offer superior milk yields, they demand intensified management including balanced nutrition, climate control, and rigorous health monitoring. "Exotic blood level alone cannot guarantee success; intensified management is the key to unlocking its potential.

Table 7: Field-level Dairy Cow blood level methods & comparison

Production System	Exotic Blood Level	Milk Yield Potential	Management Demand	Risk Level
Smallholder/low-input	50–62.5% (e.g., F2/F3 crosses)	5–8L/day	Low	Low
Smallholder/moderate-input	~62.5–67.5% (e.g., F2 crosses)	8–12L/day	Moderate	Medium
Moderate-Input	~75% (e.g., F3 crosses)	12–18L/day	Optimizes productivity while retaining some resilience	High
Commercial/High-Input	≥75 % (near-pure exotic)	15–25L/day	Maximizes yields under controlled nutrition and healthcare	High

EIAR and FAO, 2023

3.2. Development of own dairy cattle herd

Progressive development of improved dairy herds through strategic crossbreeding by artificially inseminating indigenous cows/heifers with semen from high-genetic-merit exotic bulls (Holstein/Jersey) is possible. This approach enables gradual genetic upgrading: first to 50% exotic crosses (F1-generation) yielding 6-8L/day, then through successive generations to 75-87% exotic bloodlines with an intensive level of management reaching 12-25L/day. Critically, this transformation requires theoretical & practical application of good dairy husbandry practices: preparing elite local heifers/cows, basics on dairy animal management, access to quality semen, and guidance on improved calf-rearing practices. By retaining and breeding the best female offspring, smallholders can cost-effectively build a productive herd while maintaining the hardiness of local genetics sustainable model for low-input, high-impact dairy development cross Ethiopia's smallholder farms.

3.3. Heat Detection and Estrus Management

Optimal estrus detection and reproductive management are essential components of good dairy management practices. Detecting heat (estrus) accurately and breeding at the right time boosts milk production, keeps cows healthy, and improves pregnancy rates. The average heat interval is 21 days with a range of 18 to 24 days. Making reference to reproductive records is very important, especially for a large herd.

Table 8: Stages and signs of heat

Early heat	Standing heat	After heat
• Nervousness/ restlessness. • Mounting other cows. • Swollen vulva. • Licking other cows • Sniffing other cows and being sniffed. • Reduced feed intake and isolation from the herd	• Standing to be mounted, Sharp decline in milk production (most important sign!) • Tail bent away from the vulva, nervous and excitable • The animal may stop eating • The lips of the vulva become red and swollen. • A thin, clear discharge from the vulva opening can be seen, sometimes sticking to the tail and skin surroundings	• Dried mucus on tail. • Roughened tail head • The animal refuses to be mounted • Streaks of saliva or signs of leaking on her flanks.
Early signs: Watch the cow closely	**Best signs: Take the cow for service**	**Late signs: Keep a record**

3.4. Optimal Breeding and Time of Insemination

For natural heats, artificial insemination should be performed 6 -18 hours after the first observed standing heat to align with ovulation and maximize fertilization success. Strict adherence to Artificial Insemination (AI) protocols is critical: trained personnel must follow proper techniques, source high-quality semen from genetically proven bulls, and maintain correct semen handling procedures. Progesterone testing through milk or blood samples can provide additional precision in ovulation timing. In synchronized breeding programs (Ovsynch, Pre-Synch, or CIDR protocols), strict adherence to the hormonal treatment schedule is critical, particularly for Timed AI (TAI), where predetermined intervals between hormone injections dictate the exact insemination window. This disciplined approach to timing, whether for natural or synchronized cycles, significantly improves conception rates and reproductive efficiency in dairy herds.

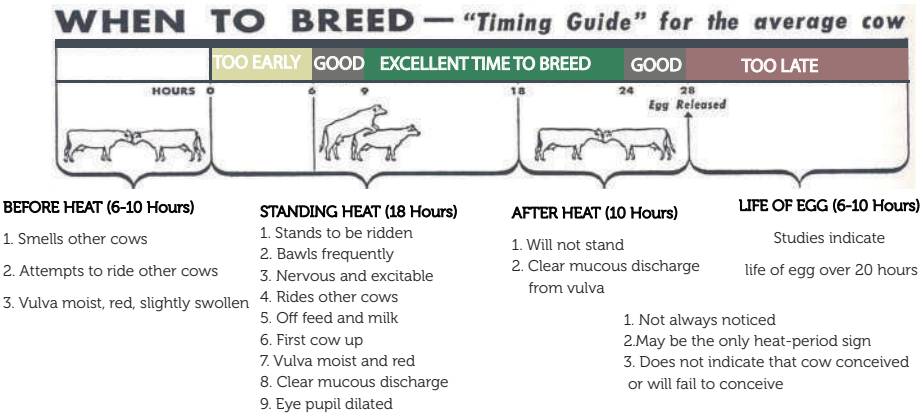


Figure 13. Decision for the best breeding time in cattle

Principle of Artificial Insemination (AI)

The biological goal is to deposit the thawed semen directly into the body of the uterus, just past the internal opening of the cervix. This targeted delivery significantly reduces the distance sperm must travel to reach the oviducts.

The technical execution of AI follows a strict sequence of core procedures known as the recto-vaginal technique. First, the frozen straw is removed from the tank and thawed in a warm water bath (34–35°C) for 15-30 seconds to restore sperm activity, then dried and loaded into a specialized AI gun. The technician then inserts one gloved, lubricated hand into the cow's rectum to stabilize and manipulate the cervix through the rectal wall. Simultaneously, the AI gun is inserted through the vagina and carefully threaded through the rings of the cervix. The final and most critical step is the deposition of semen directly into the body of the uterus, just past the internal cervical opening.

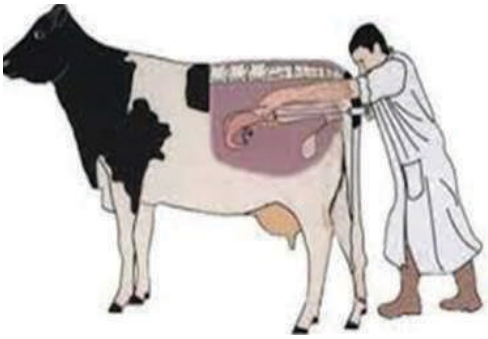




Figure 14. Site of semen deposition & critical materials for proper AI

Fertility benchmarks: The conception rate (50-70% target for first AI) directly reflects reproductive program efficacy. A conception rate below 50% signals potential issues of heat detection, timing, semen quality, or health issues.

- Days Open: <85 (calving-to-conception)
- Calving Interval: 12 months
- Services/Conception: ≤2 Services per conception
- If Conception rate values are below 50%? Check:
  - Heat detection accuracy
  - Insemination timing (6-18 hrs after heat)
  - Semen quality and handling
  - Cow health (uterine infections, nutrition)

3.5. Dairy Cow Pregnancy Diagnosis/Testing

Field-level Dairy cow pregnancy testing is essential for efficient reproductive management in dairy farming. Early detection (3 weeks for ultrasound and 60 days with manual palpation) improves reproductive efficiency, reducing feed costs and calving intervals. Cost-effective methods like rectal palpation or milk progesterone tests help cull non-pregnant cows faster, saving daily feed costs and shortening calving intervals. Reliable testing boosts milk yields and farm profitability by ensuring timely rebreeding and efficient resource use. The decision to use the test method depends on various scenarios: accuracy, affordability, practicality as well as the herd size. Here are the common pregnancy test methods used in dairy cattle. (Eg. Ultrasound for accuracy, milk tests for affordability, palpation for practicality.

For your information: Participants of Practical Guideline for Application of Reproductive Biotechnology in Dairy Cattle document preparation (page 12)

Table 9: Field-level dairy cow pregnancy testing methods & comparison

Method	Earliest detection	Accuracy	Expertise Needed	Best For
Rectal Palpation	60–90 days	~95% (after 60d)	Veterinarian/experienced tech	Smallholders
Ultrasound	28–30 days	98–99%	Trained operator	Commercial farms
Blood Test (PAGs)	21–28 days	85–90%	Lab access	Large-scale screening
Milk Progesterone	21–28 days	85–90%	Minimal training	Routine herd checks

FAO 2023, Journal of Dairy Science



Figure 15. Field-level Pregnancy Diagnosis using Ultrasound

3.6. Pregnancy and Parturition Management in Dairy Cows

A dairy cow's pregnancy lasts ~280 days (9 months) and is divided into three key phases, requires tailored care across three phases to ensure both cow and calf health. Proactive management: nutrition, vaccination, and low-stress environments-across pregnancy phases ensures healthy calves, optimal milk production, and cow longevity. The probable date of giving birth can be known if the date of service is accurately recorded.

1) Pregnancy phases management for dairy cows

I. Early pregnancy (0-3 months)

During early pregnancy (0-3 months): Pregnant cows should be treated carefully. The early stage and the later one-third of the period are very critical in the life of the foetus and the dam. In the early stages of pregnancy disturbances can cause abortion. Improve embryo viability through stress reduction, balanced nutrition, health care and pregnancy confirmation are critical during this phase to prevent early losses.

II. Mid-pregnancy (4-6 months),

In mid-pregnancy (4-6 months), priority shifts to fetal development and lactation preparation. Adjust feed to boost energy and minerals (calcium, phosphorus), while maintaining a body condition score of 3.0-3.5. Preventive measures like hoof trimming and deworming help avoid lameness and nutrient competition, ensuring steady growth.

III. Late Pregnancy (7–9 Months | "Transition Period")

The transition period (7-9 months) demands intensive management as 70% of fetal growth occurs and the dam instinctively becomes slow, gentle and separates from the herd. Key actions include feeding adjustments, dry-off 60 days pre-calving, and moving cows to clean calving house. At calving, monitor for dystocia, ensure prompt colostrum intake and check for milk fever or uterine issues. Consistent, proactive care across all phases optimizes cow longevity and herd productivity. . Knowledge of signs of approaching parturition is beneficial for the farmer to plan his operation and put the animal separately from the herd.

2) Separation & Maternity Pen Care for Dairy Cows

During the transition period (last 3 weeks pre-calving), provide a low-stress environment with a balanced diet and control calcium to prevent metabolic disorders. In the maternity pen, ensure a clean, dry, and well-bedded space with minimal disturbances. Monitor closely for calving difficulties (dystocia) and assist if needed. After birth, prioritize colostrum feeding and check the cow for milk fever, retained placenta, or infections. Allow the cow to show natural behaviour as much as possible. Proper care during this phase ensures healthy calves and optimal postpartum recovery for the cow.

3) Parturition in a cow

Parturition is the process of giving birth and it occurs at the end of the gestation period/ pregnancy. It's a very critical management phase in the production cycle of dairy cattle.

Signs of parturition: The abdomen increases in size, udder gets larger and may become firm and full, relaxation of the pelvic structure and the cervix, the cow may stand apart from the rest of the herd and becomes restless, vulva swells and discharge of mucus is normal in this stage, cow's tail may be held high and the back will be arched., Short before calving down there is relaxation of ligaments at both sides of the tail and When the udder runs full and the tail is 'loose' the moment is near.



Figure 16. Sign of pre-calving

4) Stages of calving

There are three stages during parturition of the calf. These are:-

Stage 1: Dilation of the cervix

During this stage the water bag appears filled with amniotic fluid which assists to dilate and lubricate the passage way through which the calf will pass. The cervix dilates because of the release of a hormone oxytocin and the pressure of the water bag against it. The feet of the calf should appear shortly after the appearance of water bladder.

Stage 2: Delivery of the calf

This stage is characterized by the progression of the calf through the birth canal and its expulsion. At this stage the calf may still be enclosed in the second amniotic fluid. After the head has passed through the birth canal the rest of the body usually demands little extra effort to be expelled. A common mistake is to attempt to assist by pulling on the forelegs of the calf unnecessarily or too early.

Stage 3: Expulsion of the placenta

During this stage the placenta or afterbirth is expelled from the uterus. After the delivery of the calf uterine contractions continue for a period of time. These contractions help to break down the cotyledons by separating the placenta from the uterine cornices. Normally, the afterbirth should be expelled within 12 hours of birth.



Figure 17. Delivery sign

5) Care of the Dam at parturition

Calving is a natural process for the cow/heifer. In most cases assistance is not needed and should be avoided. During a normal unassisted calving two feet should appear followed by the nose and within an hour the whole head with in minutes the shoulders and finally the entire calf will quickly be expelled. If the animal is obviously strain¬ing fruitlessly or there is any doubt of abnormal presentation immediate veterinary aid is needed. Similarly watch for expulsion of after birth (placenta), with the expul¬sion of placenta parturition is complete.

6) Care of the New-born calf

Care of the calf start before its birth. The pregnant cow should be dried off 6-8 weeks prior to expected calving date and should be fed well; otherwise, the calves born would not be as healthy as expected. The first few weeks are critical in the life of a calf care in sanitation, health and feeding. Hence the following should be addressed: -

- After giving birth, the dam licks the newborn dry and removes the mucus.
- Normally, the cow licks the calf immediately after birth. If the cow does not lick the calf, remove mucus from the nostrils and mouth of the newborn and wipe it dry with a clean towel. This act facilitates drying off the calf and helps in stimulating breathing and circulation.
- Remove the wet bedding from the pen to keep it clean and dry.
- Wash/clean the cow's udder and teats to avoid infecting the calf.
- The calf should be standing up within an hour. If not, help the calf stand up.

4. Dairy Cattle Housing and Manure Management

A proper housing system contributes significantly to optimizing animal management, farm productivity, milk quality and farm profitability. It should provide a hygienic, comfortable, protect from predators, efficient and economical labour use, and a suitable environment for hygienic milk production. Shelter enhances the comfort of the dairy animals and contributes to their good health. Without proper housing, animals face environmental stress that harms their health, production, and reproduction.

4.1. Purposes, Site selection and building of dairy cattle housing

A. The purposes of dairy cattle housing

- **Protection and Comfort:** Proper housing protects animals from harsh weather and predators while providing comfortable flooring, bedding, and space that improves overall well-being and productivity.
- **Health and Hygiene:** Good housing supports disease prevention and reduces mortality through proper drainage, waste management, and cleanliness, minimizing infections like mastitis and lameness.
- **Efficient Management**: Controlled environments make it easier to monitor health, manage breeding, detect heat, and ensure safe calving and maternity care.
- **Nutrition and Feeding:** Organized housing allows for regular feeding and watering to enhance animal health and milk production.

B. Site selection of dairy housing

Before establishing a dairy farm, it's essential to consider the farm size and expected economic returns. It's recommended to start with a small herd and a low-cost structure. Housing should be well-located, properly constructed, spaced and grouped. Ideally, when selecting a dairy farm site the following criteria should be considered:

- **Nearby market**: The nearer the market, the lesser the transportation charges and the lesser the loss due to spoilage of products during storage and transport.
- **Water supply:** Water should be available cheaply and in plenty.
- **Topography:** The topography should be high and level with no abrupt slopes. A relatively level area requires less site preparation, thus lowering building costs.
- **Drainage:** The soil should be porous and the slope gentle so that drainage is efficient and the farm premises remain dry.

- **Land requirement:** The area for the farm should be adequate size of animals
- **Sun exposure and wind protection:** The farmstead should be located in an east-to-west direction to minimize sun exposure and protection from strong prevailing winds.
- Enough electricity availability and transportation facilities.
- Telephone line facility, a school near the farm, and a Shopping center should be near
- The location should be far enough distant from other farms and residential and pollution areas.

C. Building of dairy housing

The following are strongly recommended as considerations during the construction of a dairy house:

- The shed should be built with the open side of the shed facing away from the direction of the prevailing winds.
- It should be well-drained, preferably with a gentle slope,
- The shed should not occupy good-quality land, which could be used for cultivation
- The roof may be flat or pitched (double or single pitch). In areas where winds are strong, double-pitched roofs must be used.
- The shed should be well-ventilated but draught-free, and air inlets should be sited above the level of the animals.
- There should be enough space between blocks with the Farm
- In general, it should be considering this all designing for flexibility, width of building, length of building, height of building, shape of the roof, structural form, designing for unit and orientation of shed.
- The design and construction of the house should be under the supervision of livestock professionals.

4.2. Smallholder Dairy Housing System

Dairy houses in smallholder farms are typically built using locally available materials such as timber, mud bricks, and bamboo. Roofs may be covered with grass thatch as a cost-effective alternative to corrugated iron. Floors are often made of compacted earth, sometimes with straw bedding, avoiding the need for costly concrete. Housing design and materials vary depending on agro-ecological conditions and resource availability across regions. In hot lowland areas, open-sided structures support ventilation, while in cold highland zones, walls

are enclosed with wood and mud plastering, incorporating ventilation to protect cattle from wind and cold. There are several initiatives underway to enhance smallholder dairy producers' capacity and boost milk production by promoting improved husbandry practices. These efforts focus on improving the design and layout of dairy housing systems, strengthening overall farm management, and reforming extension service delivery. Key strategies include the introduction of dairy village models and the establishment of urban dairy clusters, both aimed at fostering knowledge-sharing, resource pooling, and efficient service delivery among smallholder and urban dairy farmers

4.3. Commercial and Cluster Dairy Housing

Commercial and cluster dairy housing involves a structured and planned approach to housing dairy cattle in systems that aim to enhance productivity, animal welfare, resource efficiency, and management effectiveness.

1. **Commercial dairy farm housing:** Profit-oriented dairy farms are equipped with modern facilities such as individual stalls, proper ventilation, feeding troughs, waste management systems, and improved milking systems and etc.
2. **Cluster dairy farm housing:** A dairy farm cluster is a grouping of multiple dairy producers within a specific area to share resources, knowledge, improve productivity, efficiency, opportunities, sustainability, and innovation, and gain economic or operational advantages.

Table 10: Distance within and between commercial and cluster dairy farms in the Ethiopian context

Structure	Recommended Distance	Rationale	Remark
Residential areas (human housing)	≥ 1000 meters	Reduce odour, noise, and disease transmission	
Water sources (rivers, wells)	≥ 50 meters	Prevent contamination from manure and waste runoff	
Feed storage	≤ 20 meters	Proximity for feeding efficiency, but protected from moisture	
Manure pits	≥ 50 meters	Minimize pollution and disease risk	
Roads/access routes	≥15 meters	For milk transport and input delivery	
Public institutions (schools, health centers)	≥ 1000 meters	Public health and nuisance considerations	
Other livestock farms or clusters	≥ 3000 meters	Enhances biosecurity by limiting disease spread.	
Spacing Between Blocks of Dairy Farms (dairy farm clustering)	≥10 meters	Prevent disease spread and ensure ventilation	Dairy cluster farming

FAO, 2020, and MoA

4.4. Basic Principles and Structure of Dairy Cattle House

4.4.1. Basic principles of the dairy cattle house

These are the key aspects of dairy housing: Air (ventilation), space, lighting, rest (comfort), feeding and watering, which are critical for creating a productive environment for dairy cattle.

Air (Ventilation): Proper ventilation ensures the dairy barn is free from harmful gases (like ammonia), moisture, and excess heat. Good airflow helps prevent respiratory diseases, reduces the accumulation of harmful gases, and maintains optimal temperature and humidity levels. Poorly ventilated houses can lead to poor air quality, causing stress, reduced milk production, and potential respiratory issues in cows.

B. Space and Layout: Adequate space for each cow, calf, and heifer reduces stress, prevents injuries and promotes social interaction. The layout should allow easy movement of dairy herds and facilitate routine management activities (milking, growing, feeding, cleaning, etc..). Poor layout can lead to increased competition for resources, aggression and higher disease transmission rates.

C. Feeding Systems: Properly designed feeding systems are critical for maintaining optimal cattle health and production. An efficient feeding system helps to reduce feed waste and ensure cows receive adequate nutrition.

D. Watering: Availability of clean water for 24 hours.

E. Lighting: Proper lighting, especially natural light, helps regulate the calf, heifers, and cows' circadian rhythms, enhancing milk production. Inadequate lighting can disrupt cow behaviour and biological functions, leading to decreased feed intake, reduced growth, milk yield, and lower overall health.

F. Rest: Comfortable resting areas with adequate bedding (straw, sawdust, or rubber mats) promote cow well-being. Cows spend a significant amount of time lying down for at least 10 hours per day. Uuncomfortable resting areas can lead to joint and hoof problems, poor sleep, reduced milk production, slow growth rate, and no or poor heat detection.

4.4.2. Structure of the dairy cattle house

The farmstead structure of dairy housing refers to the organized layout and arrangement of various facilities like floor, wall, roof, feed and water trough, gutter and walkway, and doors:

1) **Floor:** The shed floor should be made from an impervious material that can be easily kept clean and dry

The floor should be non-slippery with a rough finish, but it should not be too uneven. Example grooved/corrugated cement concrete floor.

- The floor should be a slight slope (1–2%) toward the gutter for efficient drainage.
- It should facilitate efficient manure and urine removal to maintain hygiene.

2) Wall: The walls of the dairy house should be constructed with smooth finishes to minimize the accumulation of dust and moisture. The height of the wall can range from 4 to 5 meters, depending on the local climate conditions. It requires at least about 0.5 to 1m of air space to ensure ventilation is considered the most desirable.

3) Roof: The shed of the roof can be made from corrugated iron sheets. There should be a hangover of 0.5 to 1m beyond the wall to prevent rainwater from entering the cow shed.

4 Feed-through: A cement concrete feed-through is recommended for its durability and ease of cleaning. The ideal height is 45–50 cm at the front and 50–55 cm at the back to prevent cows from jumping over. A width of 60–70 cm and a length of 50–60 cm per animal is generally sufficient. For watering, provide separate spaces between adjacent feed-throughs equal in size to the feed-troughs.

5) Walkway: The central passage should be 1.5 to 2m wide (excluding gutters) when cows face outward, and 2 to 3m wide when they face inward. It should also have a slope of 1–2 % from the center toward each gutter to ensure proper drainage

6) Gutter: The manure gutter should be wide enough to hold all dung without blockage and easy to clean. It should have a slope of 1 to 2 cm per 220 cm to ensure smooth liquid waste flow. Recommended dimensions are 40 cm wide and 6–10 cm deep, with a crossfall sloping away from the standing area. Dung will be manually removed by the farmer or laborer.

7) Doors: The doors of a single-row cowshed should be 2 to 3 meters wide, and those of a double-row cowshed 4 to 5 meters wide. In both cases, the height should be 3 to 4 meters. All doors should lie flat against the external wall when fully opened.

4.5. Type of dairy cattle housing systems

Dairy housing systems refer to the various types of structures and environments used to house and manage dairy cattle. These systems are designed to provide comfort, protection, and easy access to feed and water for the cows, while also facilitating efficient milk

production, waste management, and labor use. The choice of a housing system depends on factors such as climate, herd size, labor availability, and budget.

4.5.1. Loose housing system (Open Housing)

In the loose housing system, animals are kept in groups in an open area both day and night, except during milking, treatment, or breeding. The layout includes an open space and a resting area, with a covered shed to protect animals from heat, cold, or rain. Feed and water are provided in common troughs. The entire area is enclosed with a 3-meter-high fence or compound wall, which can be adapted with minor modifications. This system is preferable for smallholder farmers and it is mostly ideal for lowland and rural areas.



Figure 18. Loose Housing System (Open Housing)

Benefits of the loose housing system: The loose housing system is cost-effective, easy to maintain, and offers greater comfort to animals. It allows for easier oestrus detection and improves labour efficiency compared to conventional systems. However, poor management can increase the risk of injury and disease spread.

4.5.2. Conventional housing (Barn) System

In this housing system, animals are tied in stalls for feeding, watering, milking, and resting within a fully roofed shed with windows or ventilation. Feed is provided in individual troughs, and milking can be done in a milking barn. The shed layout depends on the number of animals: a single-row system is preferable for up to 10 animals, while more than 10 require a double-row system. Typically, one shed holds 50–60 cattle, though modern housing systems can hold up to 100. In double-row housing, animals may be arranged tail-to-tail or head-to-head.

Tail-to-tail housing system: In this housing system, dairy animals are tied in opposite directions, making shed cleaning and milking easier while reducing disease transmission. It also allows better monitoring of the animal's hindquarters, ensures direct fresh air, and facilitates effective supervision during milking.



Figure 19. Tail-to-tail housing system

b) Head-to-head housing system: In this housing system, animals are tied facing each other. This makes it easy for the animals to go to their stalls, and feed with ease, thus enabling better management. The morning sunlight falls maximum on the gutter, which improves the overall hygiene of the shed.

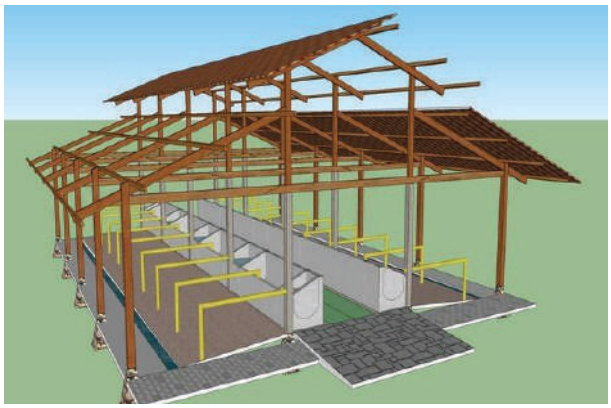




Figure 20 Head-to-head housing system

Table 11. Suggested floor space requirement for loose housing and Barn Covered Area

Types of animals		Floor space required (m2)			
		Barn Covered Area (m²/animal)		Open Area (loose house) (m²/animal)	
Calves (up to 6 months)		1.1 X 1.5	1.5–2.0	1.5 X 2.0	2.0–3.0
Growing Heifers (6–12 months)		1.2 X 1.8	2.0–3.0	1.5 X 2.5	3.0–4.0
Large heifers		1.2 X 2.2	3.0–4.0	1.5 X 3.0	4.0–5.0
Cows	Crossbred (HF)	1.2 X 2.2	3.0–4.0	2.0 X 3.4	6.0–7.0
	Jerseys	1.2 X 2.0	2.5 - 3.0	1.5 X 3.0	3.0-4.0
Maternity Pen		3 X4	12	3 X4	12
Bullpen		3 X4	12	3 X4	12
Isolation pen		3X4	12	3X4	12

Source: MoARD, 2007 and ILRI, 2010

4.6. Dairy Cattle Farmstead structure and Manure management

4.6.1. Design and layout of the dairy cattle farmstead structures

The proper design and layout of dairy farmstead structures are essential for ensuring optimal animal health, productivity, labour efficiency, biosecurity, and environmental sustainability. A well-planned farmstead, the dairy house can make it comfortable for animals and make management easier.

- The layout should depend on the number of animals to be housed, facilities to be provided for feeding, collection and disposal of manure, cleaning and washing.
- Store-houses, hay stacks, silo pits and manure pits should be located conveniently so that transportation of materials is minimized.
- For a commercial dairy farm, depending upon the climatic zones and the number of cows, the shed should be planned as a single row or double row.
- In hot, humid regions, it is advantageous to have single-row housing if the number of herd animals is not too large. If the number of animals is large, double housing with the cows facing outside (tail-to-tail arrangements) is recommended.

4.6.2. Main building partition of dairy cattle house

Dairy farms vary in scale, layout, and design based on factors like herd size, geography, climate, and available capital. However, most dairy farms share core components that must be carefully planned for efficient, hygienic, and cost-effective operations. The most important parameters for the design of dairy cattle barns are as follows:

Resting Area (Cubicles) : Each cow needs to have her own place in the resting area, called a resting area(cubicle). It should construct a number of resting areas enough to be occupied by animals most of the time. It is recommended that measurements for a cubicle be based on pure-breed cows (Friesians and Ayrshires) because pure breeds give the highest economic returns in a zero-grazing system.

A resting area has a length of 220 cm and a width of 120 cm. The resting area is separated from each other by two poles.

- If the cows are small in size (e.g., Jerseys), the measurements can also be smaller (200 x 110 cm). However, it is much better to use the above standard measurements for a cubicle because the size fits most animals.
- The standard cubicle can also be adjusted to fit smaller breeds whenever necessary. Cubicles for young animals should be adjusted according to their size by moving the neck-poles backward to make the cubicles shorter.

B. The walking area: The walking area is about 2 to 3m wide. No roofing is required for this area except under very hot conditions. The floor of the working area is made of concrete if it is affordable. This makes it possible to collect urine and manure. The floor and the cows are also kept clean.

- The surface of the floor should be rough so that animals cannot slip on it, and it should be easy to clean.
- The floor should slope from the milking place towards the urine and dung collecting pit located at one end of the walking area.
- The slope should allow for easy cleaning of the milking and walking area.

C. The milking Area (places): The milking place should be constructed next to the resting area(cubicles).

- It should also be of the same dimensions as cubicles, i.e., 120 cm by 220 cm standing cows.
- The floor should be flat and made of firm concrete and slope towards the walking area.
- The direction of the slope of the floor would ensure that dirt collected from the floor can flow through the walking area onto the manure pit.
- It should be a feed trough at the head of a milking place for feeding the cows during milking.
- The milking place should be kept clean, and noise during milking may disturb the cow, making it hold back some of her milk. For a zebu cow, a calf pen should be constructed next to the milking place. This is because most zebu cows let the milk down only when they see their calves.
- The milking area should be constructed in or outside of the housing of the milking cows' lives.

D. **Calf pen:** Calf pens are typically placed in a roofed and walled section of the farm, either near the resting area or opposite the milking space.

- Pens can be permanent or movable, and in cases of larger herds, may be built separately from adult cows.
- They should face away from prevailing winds but allow for good air circulation. Materials often include locally available wood, steel, or concrete. Fixed pens should have concrete or wooden-slatted floors with bedding, while movable ones can have raised-slatted or well-drained earth floors.
- Roofs are usually made from corrugated iron sheets or straw. Design of calf pen follows table 10.
- The space per calf 2m2 groups of calf, Proper draining or bedding, Proper lighting (artificial or natural), Proper ventilation; be at least 1 foot from the ground, and enough strength against predator invasion, Ease of cleaning, especially when a new one is introduced.
- The front of the pen should be made so that the calf can be fed milk, concentrates, and water easily from buckets or a trough fixed to the outside of the pen, and so that the calf can be moved out of the pen without lifting

E. **Heifer housing:** Well-grown dairy calves and heifers play an important role in the future success of all dairy farms.

- Determining the suitability of a barn relevant to housing heifers has three basic components.
  - First, you need to ensure that the barn is structurally sound for heifers.
  - The dimension for a heifer is suggested to be the growing heifers (6-12 months): 2.0-3.0m²) and larger heifers (>12 months): 3.0 - 4.0m2).

  To achieve management efficiency and accessibility heifer barn should be designed for ease of management, allowing caretakers to easily access feeding areas, clean walkways, and monitor the health of each heifer.
  - Good lighting, drainage, and ventilation systems contribute significantly to maintaining a healthy environment.

F. **Maternity Pen:** Maternity pens should provide cows with a quiet, isolated, and clean environment for calving.

- Each pen must be well-bedded, dry, and regularly cleaned to minimize disease risks to both the cow and calf.
- Individual pens are recommended to ensure undisturbed calving and timely colostrum intake.
- Pregnant cows should be moved into maternity pens 2–3 weeks before calving.
- The number of pens required is about 5% of the breedable female stock, and they should be located close to the farmer's quarters or milking barn for regular monitoring.
- The Design of maternity pen in each pen should include a a 3m x4m covered area.

G. **Bullpen:** It is recommended that a bullpen-

- Sited where the bull will be in touch with normal farmyard activities, including other stock movements, both by sight and smell,
- Should be large enough and strong enough to contain the animal, allow the bull to be fed and watered from outside the pen,
- Should allow the bull to be cleaned out or bedded without the hazard of direct physical contact.
- Should be safety signs warning of the presence of a bull at the entrance to any building where the bull is kept.
- The recommended dimensions of a bullpen are 3m x4m covered area.

H. Isolation pen for sick animals: The isolation of animals allows sick animals to rest and recover and prevents disease from spreading around the pen, group, or wider farm.

- It should create isolation areas for sick animals so they have little or no contact with the rest of the herd.
- Before introducing new animals, have a plan to isolate and limit direct contact with the home herd for a minimum of two weeks.
- The proposed dimension of each sick animal shall be about 3 m x 4 m(12m2) for the covered area

Holding area: Integrated cattle handling facilities, including a crush or head bail, are essential for safely restraining cattle when carrying out herd health procedures.

- The design must ensure that cattle are held securely, and it is particularly important for large herds in which large numbers of cows are treated for lameness or calving difficulties.
- The dimensions for the holding area shall be managed considering the number of animals.

J. Exercise area: The exercise area is an essential component of any dairy farm, as it helps prevent physical weakness, lameness, and other health issues in animals. Its support of this, a communal exercise area is planned, providing each animal with 6m² of space. This area also accommodates feed and a feed trough, ensuring both movement and access to nutrition are addressed within the same space.

K. Stores (feed and equipment): Under the small-scale and medium-scale dairy farms, some amount of feed is assumed to be fulfilled from the grazing. On the other hand, the farm owner is expected to start roughage feeds and concentrate feeds.

- The roughage feeds could be crop-residue or grass hays, whereas the concentrate feeds are those homemade or purchased from commercial feed processors.
- These feeds have to be protected from direct sunlight as well as from rain. Hence, a small-scale and medium-scale farm component for the storage of hay and concentrate, taking into account the number of animals under this, is suggested.
- In this way, concentrates, minerals, milk utensils, and other small equipment can be stored near the animals.
- A store is optional where finances are inadequate.
- It can, however, be built later on when funds are available. Inputs like dairy meal and drugs are kept in the living house before a store is built.
- When building a store, rainwater collection facilities and cemented floors are optional.
- The dimension for the hay storage is suggested to be 400 cm in length, 300 cm in width, with 500 cm in height. But, it depends onthe number of dairy herd.
- The dimension for the concentrate feed storage is suggested to be 300 cm in length by 300 cm in width by 300 cm in height. But, it depends on the number of dairy herd.

L. **Feed and water trough:** For optimum output from animals, adequate space is required so that animals have free access to feed and water.

- The water troughs are located near the feeding area and seldom in the resting area.
- Feed and water troughs are constructed with reinforced cement concrete.
- The feed troughs need to have smooth surfaces for easy cleaning and easy intake of feed by the animals.

Table 12: Feed and water trough, space requirement per animal

Category	Space per animal	Width	Height from floor	Depth
Adult cattle	60-75	60	45-50	40
Young calves	40-50	40	20	15

Note: The inside measurements of the feed trough should be 60 cm wide at the bottom and should be raised at least 15 cm above the ground level of the walking area. The feed trough should not be too deep or cows will have difficulty reaching feed at the bottom. If it is too shallow, spillage and wastage of feed will occur. The trough is constructed such that its outside is raised to prevent spillage of feed.

4.6.3. Dairy House Sanitation and Manure Management

Dairy housing sanitation is significantly important in terms of dairy cattle health management. The amount of dung and urine produced by animals is very variable due to differences in feed and water intakes, which in turn are strongly related to body weight and production intensity. Large amounts of dung and urine are excreted by dairy cattle due to the types and amounts of feed they feed on and their unique digestive system.

- Poor waste management can lead to the buildup of harmful bacteria, pests, and odours, negatively impacting cow health and overall farm hygiene. To address this, dairy farms should implement proper waste disposal methods for dung, urine, and leftover fodder.
- Proper waste management keeps the farm environment clean and supports animal health.
- Additionally, converting waste into compost, biogas, or organic fertilizer can reduce costs and generate income clean barn is critical to the health of the cattle, clean milk production, and profitability of the farm.

Manure disposal: Manure disposal area. Animals defecate approximately 8 percent of their body weight in a day. Large quantities of urine, leftover feed, fodder, bedding material, etc., are simultaneously produced in a livestock farm. Animal excreta is often mixed with straw and can be used as fertilizer for soils. Proper storage of manure is an important part of manure management. Manure from the cattle shed is removed at least twice a day. It can be disposed of either in solid or in liquid form (slurry).

Manure collection and storage bunker/pit: The handling of dairy cow manure is a very important operation of a dairy farm and requires much attention. Manure must be collected daily and put in a concrete storage facility or bunker whose design depends on the volume, usage, and destination of the manure. The most common ways of using manure in rural Ethiopia are:

A. Organic fertilizer: Manure is stored over a longer period of time in a concrete pit or bunker, from where it is distributed periodically to agricultural land as an organic fertilizer. This is the recommended practice for integrated dairy farms with sufficient land to grow their own fodder. Use is best made of a manure spreader. The frequency of manuring the land depends on the soil fertility, demand per crop, and availability of land. The duration of cropping season and rainfall patterns also determine how much manure needs to be stored over a certain period of time. In Ethiopia, this period would be between 2 and 6 months.

B. Bio-gas: The use of these biogas digesters is widespread in dairy farming systems across the country and the size ranges from 6-20 m^3 for small scale dairy farm. The biogas is used for domestic and farm purposes, such as lighting, cooking, and heating water for cleaning of the milk tank and the milking machine. It can also be used to generate electricity, although this requires a considerable extra investment. The slurry that is accumulated in the biogas digester can be used as an organic fertilizer for soil improvement and enrichment. For the construction of the biogas digester, it is recommended that the farmer contacts a certified biogas contractor.

5. Dairy Health Management and Biosecurity Measures

Animal health is the maintenance of animals' physical health to ensure their welfare and food production. In animals, health is defined as the absence of disease or infection in an animal's body and this can be shown by normal functioning with a state of normal behavior. Therefore, controlling risks and managing the impact of common transboundary and zoonotic diseases for better dairy animal health is crucial. The most common dairy cattle diseases can be grouped into bacterial, viral, fungal, parasitic, reproductive and metabolic diseases are described in table below.

Table 13: Common dairy cattle diseases and their prevention and control measures

S.no.	Disease's category	Causes	Transmission	Major signs	Prevention and control measures	Remark
1	**Bacterial diseases**					
A.	Bovine tuberculosis	It is a chronic disease of animals caused by bacteria called Mycobacterium bovis	Contact with infected animals. The usual route of infection is by inhaling infected droplets from the lungs by coughing. Calves and humans can also become infected by ingesting raw milk	severe emaciation, weakness, loss of appetite, weight loss, fluctuating fever, intermittent hacking cough, diarrhea, and large prominent lymph nodes	Use separate house for humans and animals Keep the house clean and environmental sanitation Pasteurization of milk to a sufficient temperature Conduct routine meat inspection Treatment of infected animals is rare because of the high cost and takes long time	It causes serious economic loss in dairy sector. It is a zoonosis which has serious public health impact
B	Brucellosis	It is a contagious disease caused by *Brucella* bacteria	Ingestion of contaminated feed and food, bacteria enter through mucous membranes, conjunctiva and wounds. Human can be infected with consumption of un boiled milk, undercooked meat and blood	Abortion in late, pregnancy Stillbirth (birth of weak calves) Retained foetal membrane Inflammation of testicles Arthritis due to chronic infections Reduced milk yield In bulls, epididymitis Infertility	Proper disposal of aborted foetus, Disinfection of contaminated site, Isolation of aborted animal, use separate house and keep clean, Test and cull positive animals, use pasteurized milk, Treatment is not effective in animals, Report to nearby health clinic	It is an important zoonotic disease that affect animals and humans Causes serious economic loss and health hazards in humans.
C	Anthrax	It is a per acute soil-borne caused spore forming bacterial called bacillus anthracis	Animals become infected by ingested or inhaled spores Humans can be infected with ingestion raw meat and milk from infected animals	Sudden death, high fever, tremors, depression, convulsion, respiratory distress, abdominal oedema, bloody milk, rumen atony, abortion is observed in acute and sub-acute cases. Discharge of exudative tarry blood from natural orifices	Vaccination, Proper disposal of dead animals preferably by incineration or deep burial with quick lime. Cleaning and disinfection. Treatment with appropriate antibiotics **Care:** the carcass should not be opened, since exposure to oxygen will allow the bacteria to form spores.	It is a fatal zoonotic disease affecting humans and animals which causes serious public health
D	Blackleg	It is an acute and fatal disease of dairy cattle caused by *Clostridium chauvoei.*	Infection occurs through contamination of the environment from infected faeces and decomposition of the carcass Infection occurs when animals ingest bacterial spores while grazing or infected feed	Inappetence, fever, rough coat, depression, lameness Emphysematous swelling around shoulder on heavy muscle	Regular vaccination. Treatment of sick animals with appropriate antibiotics	

2	**Viral diseases**					
A	FMD	an extremely contagious disease of dairy animals caused by a virus	transmitted by contact and through milk.	High fever (40-41°C) Dullness, in appetence, shaking and kicking of the feet Painful stomatitis Wounds or vesicles appear inside the mouth, dental pad, tongue and between the claws Drooling or abundant salivation Careful chewing and a characteristic smacking of the lips Drop in milk production	Test and quarantine Vaccination Wound dressing	Recovered animals remain carriers for up to 2 and half years
B	Lumpy Skin Disease	caused by virus and transmitted by mosquitoes and flies	Contact with infected animals, biting insects such as mosquitoes, stable flies, and ticks.	Pox like intracutaneous firm nodules swelling on the body, Tongue, gum and hard palate lesions Fever, lacrimation nasal discharge and hyper salvation followed by Swollen lymph nodes Swollen/tender udder or testicles Lowered milk production	Vaccination Slaughter infected and exposed animals Quarantine sick animals Clean and disinfect of premises. Wound disinfectants and protective dressings	
3	**Metabolic diseases**					
A	Milk Fever	a disease of high lactating cows which caused by calcium deficiency in the body of animal	Occurs just before 12 hours of calving and 24-48 hours after calving.	-loss of consciousness, dry muzzle, cold ear, pupil might be dilated, -show sitting position often and sternal recumbency with a twist or curvature of the neck, drowery appearance and flaccid paralysis. -muscle tremor, grinding of teeth, subnormal body temperature, loss of muscle	Supply more phosphorus and low calcium during late stage of pregnancy Administration of calcium chloride 120-150 ml orally 24 hours before calving Supply the cow with balanced diet during pregnancy Treatment is administration of calcium borogluconate solution	
B	Ketosis	Caused when the demand of energy exceeds energy intake.	Occurs in lactating animals usually during the first 2 months of lactation	Inappetence, drop in milk production, lethargy, weight loss, pica (animal seeks course materials), licking of bodies, head and nose pressing, chewing, bellowing, staggering, circling and falling	Proper feeding during late lactation and dry period, Avoid overfeeding and sudden change of feed	
4	**Reproductive disease of dairy animals**					
A	Mastitis	Mastitis is an inflammation of the udder/mammary gland due to infection by bacterial or mycotic pathogens	The bacteria which cause Mastitis could be present and live everywhere in the dairy farms like on the stable floor, in dung, on the skin of cattle and on the milker's hands. This leads to infection due to contact. Infection also arises due to break the teat by improper milking technique	It can be recognized by swelling, heat, pain, and abnormal secretion in the gland The clearest and often first sign of mastitis is characterized by the changes in the milk (like flakes and lumps in the milk, color change. and watery milk, instead of creamy yellow or blueish, milk can contain blood cloths and looks pink	clean milking equipment and barn, Proper washing and keeping dry hands, complete milking, disinfect teat after milking contamination, intramammary infusion of the special antibiotic tubes and Culling of chronic cases which are unresponsive to treatment	
B	Uterine prolapse	Bovine uterine prolapse is a protrusion of uterus due to severe strain and certain disorders.	It commonly occurs after calving in dairy cattle and occasionally with hypocalcemia	Protrusion of uterus mass	Place the prolapsed mass gently over a clean surface and protect it from dirt and flies Gently wash with saline solution if excessively soiled Prone animals may be kept with the hind quarter at a slight elevation	Management of the with bare hand exposed to zoonotic infection, use PPE
C	Retained placenta	Retention of fetal membranes in cows is failure to expel fetal membranes within 24 hours after parturition due to certain disorders	The case is increased in cases of abortion, dystocia, milk fever, twin birth, calving, infection and nutritional disturbances.	Hanging of placental mass	Never pull the placenta on your own. Consult a veterinarian to avoid complications Dispose the removed placenta by deep burial Avoid handling it with bare hands	Management of the with bare hand exposed to zoonotic infection. use PPE
5	**Diseases due to faulty feed**					
A	Bloat	It is a form of indigestion marked by excessive accumulation of gas in the rumen	It can occur when; the animal grazes on lush wet young plants like clover, lucerne and alfalfa Choking due to foreign objects (esophageal obstruction)	The left flank balloons out or distended The animal kicks its belly Difficulty in breathing. In severe cases, death occurs due to asphyxiation.	Avoid moving animals to wet pasture in the morning. Do not allow very hungry animals to graze a pasture. Offer dry, cut grass first before turning out to graze. In severe cases, puncturing the left flank with a trocar and cannula	

B	Rumen acidosis	is caused by ingestion of large amounts of highly fermentable, rapidly digestible carbohydrate-rich feeds	The highly fermentable, rapidly digestible carbohydrate-rich feeds which causes excessive production and accumulation of acids in the rumen.	In severe cases, prolonged recumbency Animal may not urinate or pass dung. reduced feed intake, weight loss, diarrhea and exhaustion Laminitis and lameness	Proper feeding practices, Provide a roughage diet in the mornings before feeding concentrates, Never provide large quantities of rapidly digestible carbohydrates.	
C	Laminitis	It is a disorder of the corium of the claws of cows.	It occurs mostly around calving and is strongly influenced by management factors It can be predisposed by higher concentrate/protein, low forage, infections,	Sole ulcer, laminitis, dropped sole, hoof deformity	Provide soft flooring Provide balanced feed including sufficient forage. Practice foot bath using 5% copper sulphate, Apply regular hoof trimming. Provide adequate lying space, Check hoofs regularly for any injury	
6	**Diseases of newborn**					
A	Calf scours (Diarrhea)	caused by nutritional disorders, viruses or bacteria.	Digestive upsets leading to scours are a major cause of death in young calves.	watery or loose feces, dehydration, weakness and lethargy, loss of appetite, fever and abdominal distension	ensuring calves receive adequate colostrum within 6 hours of birth and therefore acquire some natural immunity	
B	Aspiration pneumonia	Caused by fluids going to the lungs via the windpipe (trachea) during drenching	Faulty drenching of milk, when feeding rate is faster than swallowing rate	increased respiratory rate, open-mouth breathing, cyanosis (bluish discoloration of mucous membranes), fever, nasal discharge	Proper drenching and/or feeding	
7	**Parasite Infestations**					
A	External parasites	External parasites include ticks, lice, fleas and mange mites	Contact due to common grazing	Itching and scratching of affected area due to irritation, Skin Irritation and Inflammation Hair Loss or Wool Loss Visible parasite on body Formation of crusts or scabs on the skin and Anaemia	Cleaning animal housing pasture management prevent contact with another herd Application spraying of acaricide, Spray the animals once every 2 weeks for tick control	
B	Internal parasites	They are worms found mostly in the respiratory and digestive systems	Ingestion of contaminat-ed forage, Contact due to common grazing	emaciation or progressive weight loss, diarrhea rough hair coat, pot belly, anaemia (pale mucous membranes, decreased appetite, lethargy and swollen abdomen	Administration of routine deworming, Paddocking and rotational grazing, keep animals away from marshy areas and use zero grazing system	

NB: For further clarification and information, please consult the nearby animal health professionals.

5.1. Public health risks associated with dairy animals

A. Zoonotic disease transmission

- Zoonotic diseases are diseases that can be transmitted from animals to humans.
- Dairy farm workers, veterinarians and consumers are at risk for zoonotic diseases.
- The most important zoonotic diseases associated with the consumption of milk and milk products are brucellosis, tuberculosis, salmonellosis, E. coli infection, campylobacteriosis, etc.
- Zoonotic diseases can be transmitted through: contact with infected animal and their products, being bitten by infected animal, ingestion of contaminated food and water, consumption of raw meat, milk, blood, etc.

Preventive measures of zoonotic diseases:

- keep animals' health using routine vaccination and good husbandry practice,
- use separate house for humans and animals,
- consume thoroughly cooked animal products,
- keep personal health and hygiene.

B. Milk borne infections

- Milk and milk products can harbor microorganisms and be important sources of foodborne pathogens like Salmonella, Staphylococcus aureus, Campylobacter, E. coli, Listeria, etc. .
- Prevention methods of milk borne infections include
- Keep personal hygiene of milkers
- Do not consume raw milk

C. Drug residues

- The use of antimicrobials, both in humans and in animals, has driven the emergence of antimicrobial resistant (AMR) pathogens.
- After used in dairy animals, the particles of antibiotics slowly come out through milk.
- Such milk, if consumed by humans, get these antibiotics particles into their body.
- While the germs present in the human body and get exposed to these antibiotics, they might create resistant to the same antibiotics.
- If the resistant antibiotics cause disease in human body, these antibiotics will not work and fail to recover from the disease.

Preventive actions

- Do not buy and sell milk from the animals that are under treatment until the course of treatment (withdrawal time) complete.
- Suggest milk producers not to treat animals on his/her own with antimicrobials.

D. Pesticide residues

- Pesticide is a chemical substance used for destroying insects or other harmful organisms of crops/fodder plants.
- When animals take plants or feed sprayed with pesticides, the particles of the pesticide come in the milk they produce and cause toxic effects in human body.

Preventive actions include:

- Inform the milk producers to avoid using chemical pesticides in the farm
- Not provide dairy animals feed sprayed with chemical pesticides

E. Aflatoxins

- Aflatoxins are poisonous carcinogens produced by fungi, mainly Aspergillus.
- Aflatoxins can contaminate dairy feeds and cause animal health risks

- Milk gets contaminated with aflatoxins due to the consumption of moldy feed or fodder.
- When cows consume feeds containing aflatoxins (B1 and B2), the liver's metabolic process
- produces aflatoxins M1 and M2, which are then excreted in milk.

Then, Aflatoxin can be transferred from the diet to the milk of cows as aflatoxin M1 (AFM1), posing a significant health hazard to the milk consumers.

Preventive actions:

- Feed and fodder given to the animals should be stored in a clean and dry place
- Stored feed should not come in contact with water/moisture
- Storage area should have proper ventilation

F. Adulteration

- Adulteration in milk is an act of the addition of any external substance like water, skim milk powder, etc. in milk. Adulterants alter the natural quality of milk.
- Possible harmful effects on humans are:
  - Gastrointestinal distress and allergic reactions
  - Liver and kidney and cardiovascular risks
  - Create toxicity in the body
- Measures to reduce adulteration in milk & milk products are:
  - Milk producers/traders should know the harmful effects of adulterants on human health
  - Milk producers consider a moral obligation to supply quality milk for the better health and well-being of the community.
  - Efforts should be to earn more by selling more and quality milk to satisfy customers.
  - Consumers should be informed and vigilant about associated potential health risks.
  - Avoid buying milk from sellers who are perceived to compromise milk quality.

5.2. Dairy Diseases Prevention and Control Measures

Effective disease prevention and control strategies are essential for maintaining Dairy health, enhancing productivity and ensuring the welfare of animals. By implementing a multi-faceted approach, dairy producers can minimize the risk of disease outbreaks and maintain healthy flocks.

5.2.1. Dairy Biosecurity Measures

- Biosecurity refers to every measure taken to protect your herd, farm, and livelihood from the introduction and spread of infectious diseases.
- Biosecurity mean "Bio" means life and "Security" means safeguard. Thus, biosecurity mean safeguarding the life and health of animals

5.2.2. The Importance of Biosecurity

A strong biosecurity plan is the basis of a successful dairy farm. Therefore the importance of biosecurity are:

- Animal health and welfare: prevents suffering and improves quality of life for animal.
- Production efficiency: healthy cows produce more high production and quality milk.
- Farm profitability: reduces costs from treatment, production loss, and animal death.
- Public health: prevents the spread of zoonotic diseases

5.2.3. Types of Biosecurity

There are three types of biosecurity in dairy farming. These are conceptual biosecurity, structural biosecurity and operational biosecurity. These categories are classified based on how biosecurity

Table 14:Types of biosecurity and their function

Type of Biosecurity	Function	Key Measures
Conceptual	Farm location and early planning	Site selection, distance from other farms, topography, wildlife control, clean water sources
Structural	Physical layout and facilities	• Fences, zoning (Green, Yellow, Red), • Install fencing and gates to control movement • Provide footbaths, handwashing stations, and disinfection points • Construct separate housing for different animal groups (e.g., calves, sick animals) • Construct well drainage, ventilation, quarantine pens, feed and manure zones • Maintain well-drained floors and cleanable surfaces • Use dedicated equipment and storage areas
Operational	Daily practices and human/animal traffic control	• Quarantine and health checks for new or returning animals • Proper sanitation, deworming, and vaccination schedules • Control movement of people, animals, equipment, and vehicles • Monitor animal health and maintain accurate records • Educate staff and enforce compliance with biosecurity protocols • Isolation, movement control, visitor logs, pest control, disinfection, and staff hygiene

5.2.4 Operational Biosecurity (The daily practices and procedures)

- Operational biosecurity is the routine/day-to-day management practices and procedures carried out by people on the farm to prevent disease introduction and spread.
- Operation bio-security has three major components/Pillars.
- These three are isolation, Traffic Control, and Sanitation.

A. Isolation: prevents contact between animals within a controlled environment. This includes control of commingling and movement of animals (all new purchases as well as commingling between established groups of animals, segregation of sick animals, etc).

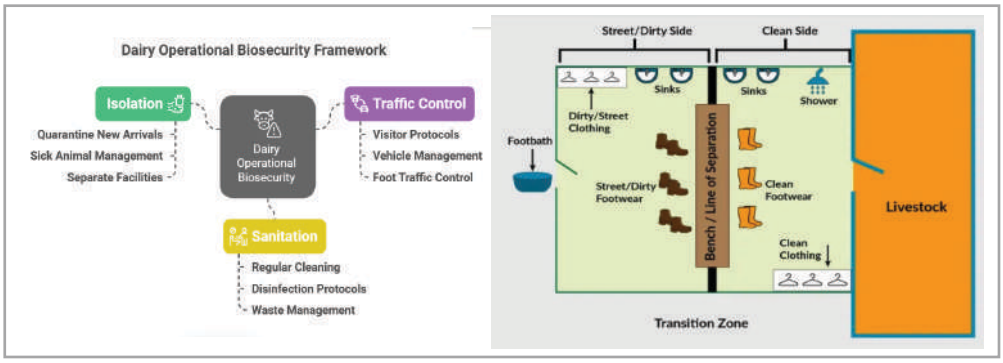


Table 15: Isolation Measures

Isolation measure	Details Activates
Animal Sourcing	• Source animals from trusted, disease-free farms only. • Request health certificates and vaccination records before purchase. • Avoid buying from livestock markets with unknown health history. • Transport animals using clean, disinfected vehicles.
Quarantine	• Quarantine all newly purchased or returning animals for at least 2 weeks. • House them in separate quarantine pens, far from the main herd. • Limit access to dedicated staff and equipment for the quarantine zone. • Monitor for clinical signs of disease before mixing with herd.
Internal Separation	• Separate age groups (calves, heifers, adults) and vulnerable animals (e.g., pregnant or sick cows). • Isolate sick animals immediately in a designated sick pen. • Avoid commingling animals from different sources or pens. • Use color-coded tools, feeding equipment, and clothing for different sections.
Health Monitoring	• Conduct daily observation for signs of illness (e.g., fever, cough, diarrhea). • Keep health and treatment records for each animal. • Screen animals regularly for common diseases (e.g., mastitis, FMD, brucellosis). • Report unusual illness or sudden death to a veterinarian promptly.

B. Traffic control measures: Traffic control includes traffic onto livestock facilities and traffic patterns within the facilities. This includes control of the movement of vehicles, animals, people, pet animals, wildlife, rodents and birds, etc.

Sanitation practices of cleaning and disinfection are intended to stop the transmission of infectious agents. It involves inactivating or destroying disease-causing microorganisms on the farm premises, equipment, vehicles and personnel.

The general order for the sanitation process is:

Table 16: Traffic Control Measures

Traffic control measures	Key Actions
Staff Movement	• Control entry/exit of people, animals, vehicles • Use Footbaths • Limit staff movement between animal groups (e.g., calves, sick cows, lactating cows). • Assign specific duties to staff to avoid cross-contamination. • Provide clean boots and clothing or disinfection stations at entry/exit points. • Train staff on biosecurity and hygiene practices.
Visitors	• Restrict non-essential visitors. • Require all visitors to sign a logbook and declare prior animal contact. • Provide protective clothing and footwear. • Designate specific clean zones for visitors; no entry to animal areas without permission.
Service Providers (veterinarians, AI technicians, etc.)	• Schedule appointments to minimize overlap and time on-site. • Ensure tools and equipment are disinfected before and after use. • Provide them with biosecurity protocols upon arrival. • Limit access only to necessary areas.
Animal Movement	• Quarantine new or returning animals for minimum 2 weeks. • Separate sick animals and restrict movement until recovery. • Use dedicated pathways and tools for different animal groups. • Clean and disinfect handling facilities regularly.
Equipment and Vehicles	• Separate clean/dirty traffic routes, • Isolate feed & manure routes

C. Sanitation Measure: Sanitation practices of cleaning and disinfection are intended to stop the transmission of infectious agents. It involves inactivating or destroying disease-causing microorganisms on the farm premises, equipment, vehicles and personnel.

The general order for the sanitation process is:

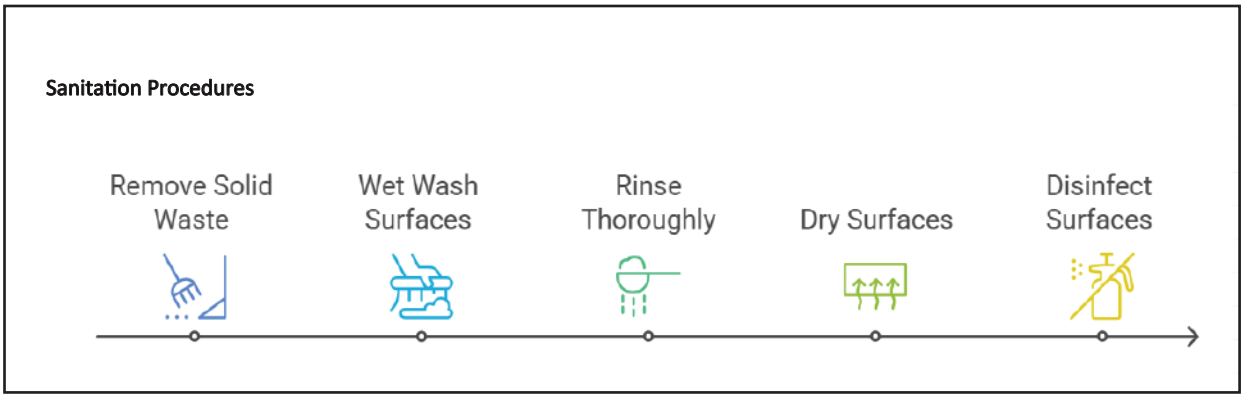


Table 17: Sanitation measure

Sanitation measures	Key Actions
People Hygiene	• Always wash hands before and after animal contact. • Use clean, farm-specific clothes and boots. • Provide handwashing stations with soap and water at farm entrances. • Avoid contact with animals without washing
Footbaths	• Place footbaths at entry points to barns and animal areas. • Use appropriate disinfectant and change daily or when dirty. • Ensure depth allows full boot coverage.
Calf Management	• Clean and disinfect calf pens before placing new calves. • Disinfect umbilical cord, • Provide clean bedding and replace regularly. • Ensure hygiene in milking and feeding • Use separate feeding tools per calf to prevent cross-infection. • Avoid overcrowding to maintain hygiene and air quality.
Milking Hygiene	• Handwashing, clean machines • Wash and dry udders before milking. • Use pre- and post-milking teat disinfectant. • Clean and sanitize milking machines before and after each session. • Milk sick animals last and disinfect equipment afterward.
Feed/Water	• Store feed in dry, rodent-proof containers. • Provide clean, fresh water daily; clean troughs regularly. • Avoid mouldy or contaminated feed
Equipment Cleaning	• Clean shared tools/equipment , • Disinfect all equipment after use, • Clean carts/tanks after use • Store clean equipment off the ground in a dry area. • Use dedicated tools for different zones (e.g., calves vs adult cows). • Train staff on proper cleaning and disinfection methods.
Waste Management	• Remove manure daily from barns and pens. • Store manure in a designated area away from feed and water. • Use proper drainage to prevent wastewater pooling. • Recycle organic waste through composting or biogas systems. • safely dispose of veterinary waste(needles, syringes)
Dead Animal Disposal	• Remove dead animals immediately from the herd area. • Manage waste and carcasses properly • Use designated disposal zones far from barns and water sources. • Apply approved disposal methods: burial, burning, or rendering. • Burying the carcass in a pit on the farm's property • Burial method: Dig a pit at least 1.5–2 meters deep • Clean disposal area • Keep detailed records of carcass disposals.

5.3. Vaccination

Vaccination is a key preventive health tool in dairy farming. It enables cows, calves, and heifers to develop antibodies against specific diseases, so if they later encounter the pathogen, their immune systems can respond effectively. Proper vaccination reduces both the likelihood and severity of disease, acting as a form of insurance to protect against potentially costly health issues

Importance of animal vaccines:

- Protection of individual animals
- Herd immunity to break chains of transmission
- Control of emerging, re-emerging and exotic diseases of animals and people
- Reduction of transmission of diseases
- Control of zoonotic diseases (e.g Anthrax)
- Reduction of animal suffering
- Reduction in the need for antibiotics
- Control diseases of animals

Table 18. Characteristics of Vaccine

Vaccine Activities	Key Points
Types of vaccine	• Live (attenuated), Example: Lumpy Skin Disease (LSD) vaccine • Killed (inactivated)Example: Foot and Mouth Disease (FMD) vaccine
Purpose of vaccination	• Prevent major infectious diseases like FMD, LSD, Brucellosis, Blackleg, Anthrax • Reduce morbidity and mortality • Improve milk production and reproductive performance • Protect calves via maternal antibodies (through colostrum) • Minimize economic loss and ensure farm sustainability • Reduce antibiotics and zoonotic risks
Precautions	• Only vaccinate healthy animals but sick animals should not be vaccinated • Follow the manufacturer's schedule and dosage • Use sterile equipment and maintain hygiene during vaccination • Record date, vaccine type, animal ID, and batch number • Avoid vaccinating during extreme weather or stress periods
Storage and Use Vaccines	• Store vaccines in a cold chain (2–8°C) – never freeze • Use a cool box when transporting vaccines to the farm • Protect vaccines from heat/light • Check expiry date before use • Shake vaccines well (if required) before administration • Discard opened vials at the end of the day

Dairy Cattle Vaccination Schedule: A dairy cattle vaccination schedule is a planned timeline that outlines which vaccines should be given to dairy animals, at what age, and how often they should be repeated.

Follow a Vaccination Schedule: develop and stick to a farm-specific vaccination calendar in consultation with a veterinarian.

Table 19: Vaccination Schedule

Vaccine	Species	Age at 1st Dose	Frequency	Notes
FMD	Cattle	4 months+	Twice a year	Based on the calendar
B. Pasteurellosis	Cattle	-	Twice a year	
Blackleg	Cattle	6 months+	Annually	
LSD	Cattle	-	Annually	
Anthrax	Cattle	-	Annually	Also zoonotic
CBPP	Cattle	-	Annually	

Source: NVI

5.4. Deworming

- Deworming is the process of administering medication (anthelmintic) to eliminate internal parasites (worms) such as roundworms, tapeworms, liver flukes,stomach and intestinal worms.
- Parasites can lead to various health issues such as weight loss, decreased milk production, and weakened immune systems
- Regular deworming is essential for maintaining the overall health of the herd, as it helps prevent parasitic infections that can reduce productivity.
- Deworming schedules vary by areas, climate, and farm management practices, but it is commonly done at the start of grazing season, before calving, and at weaning.

Table 20: Characteristics of Deworming

Deworming activities	Key points
Purpose	• Eliminate internal parasites (roundworms, tapeworms, liver flukes) • Improve feed efficiency, weight gain, and milk yield • Enhance growth in young stock • Boost immune response and reduce disease susceptibility • Improve reproductive performance in breeding animals
Timing	• Calves: First deworming at 1–2 months, then every 3 months • Adult cows: typically, twice a year • Before breeding and late pregnancy (with caution) • After the rainy season, when parasite load is high • As per the veterinarian's recommendation and faecal tests
Caution	• Avoid deworming in early pregnancy (first 45 days) – may cause abortion • Do not use expired or unapproved products • Follow the withdrawal period before using milk/meat from treated animals • Never overdose – it may cause toxicity • Consult a veterinarian for proper drug selection and dosage
Best Practices	• Rotate deworming drugs (Albendazole, Ivermectin, Fenbendazole, etc.) to avoid resistance • Deworm all animals at the same time to prevent reinfection • Use body weight estimates to determine correct dosage • Keep records of date, product used, dosage, and animal ID • Combine deworming with pasture management to break parasite cycle • Consult veterinarian, veterinarian detect/use faecal tests, monitor results, never deworm sick or stressed animals.



6. Hygienic Milk Production and Handling

6.1. Hygienic Milk Production

Hygienic milk is globally defined as milk obtained from healthy dairy animals, has its natural flavor, normal composition, free from any foreign matter and with small number of bacteria and safe for human consumption. Thus, dairy farmers must ensure that milk is produced cleanly and hygienically to improve benefits from the farm and guarantee the consumers. Milk is a very good media for microorganisms' development so it can be easily contaminated during production, handling, transporting and marketing.

Importance of Hygienic Milk Production: A hygienic and safe milk supply that will not endanger consumer health and good quality food is essential for proper nutrition. Hygienic milk production:

- Helps to improve and ensure food safety and quality
- Ensures prevention of food-borne diseases.
- Improves consumers' confidence in milk consumption
- Enhances the shelf life of milk and milk products
- Promotes participation in the international trade of milk products
- Improves farm profitability
- Improves the commercial values of milk and milk products/stimulates economic development

6.1.1. Characteristics and composition of hygienic milk

Characteristics of clean milk: Milk is a yellowish-white, non-transparent liquid with low viscosity, a slightly sweet taste and good flavour, and carries hardly any smell. Milk when it is produced from a healthy udder contains only a very few bacteria and no foreign materials. According to the Compulsory Ethiopian standard 2021, CES 278 raw whole cow milk shall be free from pathogenic microorganisms and shall comply with the microbiological limits of 1,000,000 and 50,000 cfu/ml for TBC and TCC respectively. Somatic cell count shall not exceed 300,000 cells/ml. It contains a natural inhibitory system which prevents a significant rise in the bacteria count during the first 2-3 hours. If milk is cooled within this period to 4°C, it maintains nearly its original quality.

Composition of quality milk: Milk consists of about 87% of water and 13% are nutritionally important components, such as proteins, carbohydrates, lipids, minerals and vitamins. The composition of milk is not constant but shows a wide variation. Milk composition depends on the species of the animals, within a species, between individual animals within a breed. The composition might even change from day to day depending on feeding and climatic conditions.

Table 21: Composition of milk (%) from different cattle breeds.

Breed	Fat	Protein	Lactose	Ash
Local breed/Zebu	5.6	3.1	4.6	0.71
Ayrshire	3.8	3.4	4.8	0.70
Friesian	3.4	3.2	4.6	0.74
Guernsey	4.9	3.8	4.8	0.75
Jersey	5.1	3.8	4.9	0.75
Shorthorn	3.6	3.4	4.8	0.70

Table 22: Raw whole cow milk shall comply with the compositional requirements (Ethiopian standard 2021, CES 278)

Characteristic	Requirement	Test methods
Milk fat, %, Min.	3.34	ES ISO 1211, ES ISO 2446
Solids not fat, %, Min.	8.38	AOAC 990.21
Density at 20 ^{0}C	1.026 g/mL – 1.032 g/mL	AOAC 961.07
Freezing point depression	- 0.525 0C to – 0.550 0C	ES ISO 5764
pH	6.6 – 6.8	Annex C
Total protein, %, Min.	3.07	ES ISO 8968-1, ES ISO 8968-4
Alcohol test	Negative	ES 3471
Titratable lactic acid, %, Max.	0.2	ES ISO 11869
Clot on boiling test (COB)	Not coagulate	ES 3458

6.1.2. Milking operations

Milking is the most important activity in a dairy farm. Milking can be either by hand or machine. Hand milking is an art, which is improved with practice. For efficient milking, the teat should be of moderate size, symmetrical and enough tension of the sphincter muscle.

i. Hand milking

Milking is done either by stripping or full hand methods. Below are the steps we follow during hand milking:

Step 1: Preparing milking items:

Milking bucket, Graduated jug, Milk strainer or sieve (muslin cloth), Milk cans or bulk container, Towel (2 towels/cow), Dip cups and Strip cup, Teat disinfectant, Waste basket, milker's seat and rope.

Step 2: Apply milking order

- Start milking with first calvers
- Follow by uninfected cows
- Segregate chronic mastitis cows, milk them last

Step 3: Provide well-managed and clean cows

Step 4: Wash with soap and disinfect hands before milking.

Step 5: Wash udder and dry with a clean towel

- Wipe dirt and debris from the udder
- Use more than one towel, if necessary, to dry the teats
- Completely clean teats from the base of the udder to end of the teats
- Pay special attention to the tip of the teats where the opening is located
- Washing the udder not only makes it clean but also stimulates milk let-down



Figure 21. Cleaning the udder and teats of the milking cow

Step 6: Use a strip cup and remove the fore milk into a strip cup.

- Squirt milk onto the black surface of the strip-bowel.
- Foremilk should be checked and/or examined and abnormal milk should be discarded.
- The foremilk should not be allowed to run on the floor as this increases the danger of contamination.
- Abnormal milk may change colour, appear watery, bloody, or have clots or flakes. This may indicate infections like mastitis.
- Cases of mastitis must be regularly checked and monitored by screening technique and tests such as strip cup detection (Clinical mastitis) and;
- Sub clinical mastitis can be detected by examination of the somatic cell counts using either California Mastitis Test (CMT) or Somatic Cell Counts (SCC). screening methods and test procedures indicated in Annex 2.

Step 7: Steps in hand milking

- Sit at the right side of the cow preferably and use both hands alternating during milking.
- Take hold of the base of the teat
- Squeeze with thumb and fore finger
- Close the other 3 fingers and squeeze them in turn. Rhythmically repeat this.
- Milk quickly and evenly (remember 7 minutes of let-down)

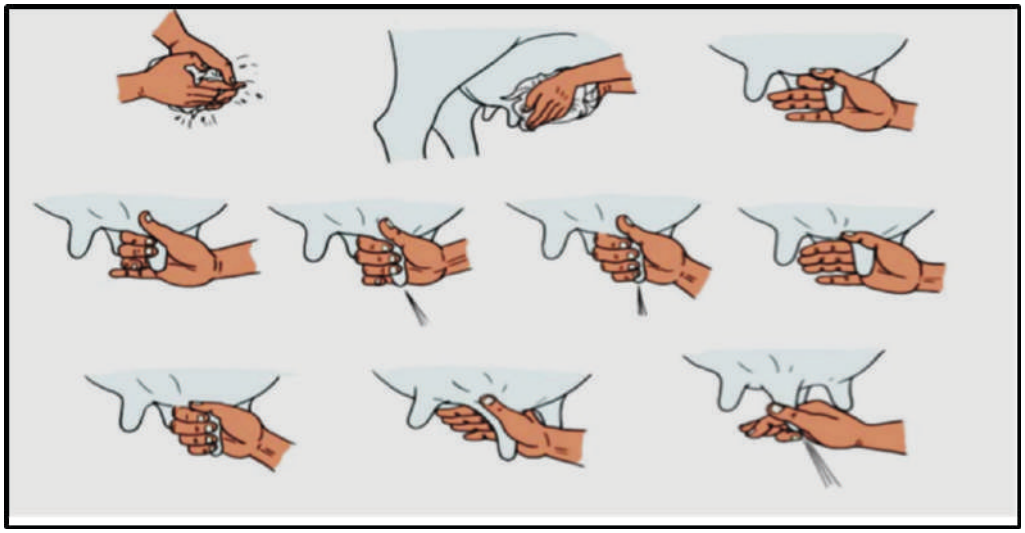


Figure 22. Step by step method of hand milking

Step 8: Dip cow teats in disinfectant

- Use teat dip (i.e. disinfectants, iodophore, etc.) after milking, to restrict entry of microbes to the teat canal and prevent infection of the udder (mastitis).
- Immerse at least 3/4 of the teats into disinfectant and allow 30-second contact time.



Figure 23. Dip cow teats in disinfectant

Note: -

- Routine milking procedures stimulate milk letdown and should therefore not be changed unnecessarily.
- After the cow has been maximally stimulated for milk let down, it should be milked immediately since the stimulus reduces over time. Oxytocin effects are maximum between 5-10 minutes; thus, milking should be completed during this time.
- Don't harass animals since adrenaline (a hormone produced due to fright) has the opposite effect of oxytocin (milk let-down hormone).
- It is recommended that the animal remain in a standing position for at least one hour to ensure the teat does not come into contact with the ground while the sphincter is still loose.

Modern milking machines are capable of milking cows quickly and efficiently without injuring the udder, if they are properly installed, maintained in excellent operating conditions, and used properly. The milking machine performs two basic functions.

- It opens the streak canal through the use of a partial vacuum, allowing the milk to flow out of the teat cistern through a line to a receiving container.
- It massages the teat, which prevents congestion of blood and lymph in the teat.

Advantages of machine milking

The advantages of this milking machine are manifold. These are;

- It is very hygienic and all the milk from the udder can be removed.
- The machine is also easily adaptable and gives a suckling feeling to the cow and avoids pain in the udder as well as leakage of milk.
- It is preferable for large-scale dairy farm milking operations.

6.1.3. Methods of improving hygienic milk production

Good hygiene and quality control need to be observed at all stages of milk production, handling and marketing. Thus, hygienic practice must begin at the farm level. Good hygiene will ensure that the milk you handle is clean and has low levels of spoilage/contaminants. Hygienic milk production can be achieved by improving cleanliness and healthy of cows, hygiene of housing and milking environment, Hygienic and healthy feed and water, personal hygiene and health of milk handlers, proper milking, and hygiene of milk utensils.

A. Clean and healthy cows

The cow is an important source of contamination. The skin of the animal provides a large surface for possible contamination. Particularly the teat and udder can be the major source of contamination of milk. Care and management practices for clean and healthy cow:

- The cow health and hygiene is the starting point for clean milk production, so the cow should be kept in good health and hygiene.
- Examine periodically for udder and other infections.
- Infected animals should be treated and isolated from the normal herd to avoid further spread of contamination.
- Cows with mastitis should be milked last and their milk should be discarded.
- The udder, teats and tail should be washed and groomed daily probably before milking so that the dirt particles in the air do not fall into milk.
- Milk of the infected animal should never be mixed with bulk milk, until the animal recovers from the illness fully.

B. Hygiene of housing and milking environment

Clean barn is critical to the health of the cattle and clean milk production. A neat and clean housing is quite important to have healthy animal that will produce hygienic milk at farm. The environment in and around the milking premises also determines to a great extent the contamination of milk. Animal house needed to be designed along the following recommendations:

- Animal stables have to be located on high ground with a natural sloppy drainageFloors must be made of washable surface, hard and easy for cleaning.
- The walls should be smooth and washable.
- Ensure proper drainage of dung and urine directly to the sewer or for frequent removal.
- Ensure proper aeration/ventilation and sunshine in shed.
- Have provision for regular supply of clean and fresh water.
- Adequate lighting is needed to carry out milking and cleaning operations satisfactorily.
- Equipment used for feeding and watering should be properly cleaned and maintained.
- The sheds should be maintained properly by:
  - Keeping the shed clean, wash regularly and dry as much as possible.
  - Freeing milk houses from dust and stable odours.
  - Removing the dung and urine frequently, away from the shed.
  - Spraying a recommended chemical in the dung pit to stop the breeding of mosquitoes.
  - Using fly repellents (i.e. phenyl) inside the shed and/ or at the farm premises.



Figure 24. Clean the animal house for clean milk production

C. Hygiene of feeding and watering

Always feed the animal with a quality feed and water, as a healthy animal will yield the clean milk. Feeding management practices for clean milk production includes:

- Feeding the animals with healthier feeds will reduce the chances of occurrence of diseases.
- Clean the water tubs and the feeding manger regularly to avoid microbial growth.
- Avoid feeding silage and hay during milking.
- Never feed the animal with leftover feed that may be spoiled with molds or other microbes.
- Never allow the animal to drink dirty water as it may lead to waterborne infections.

D. Milker hygiene and health

Milker hygiene and health practices are crucial for producing high-quality, safe milk and preventing udder infections like mastitis.

- Milkers should be free from communicable disease.
- Periodic medical checkup should be arranged at least 2 times per year.
- They should not also have open cuts including sores, boils or infected wounds.
- Milker should always wear neat and clean clothing and caps.
- Milker should also maintain adequate personal cleanliness (i.e. haircut, trimming of beard and cutting the nail regularly) to avoid any microbial contaminations.
- Wash hands with detergents and drying with clean towel is must prior to hand milking.
- Not wipe hands on the body of the animals or on his/her own body.
- Never smoke, eat drink or spit while milking.
- Avoid coughing and sneezing while milking.




Figure 25. Personal hygiene and dos and don'ts during hand milking

E. Proper milking

Milking should be carried out to minimize contamination of the milk being produced. A good milking routine prevents contamination of the milk and helps to elongate the quality and shelf life of the milk produced. During milking the following conditions should be considered:

- Correct milking practices should be adopted to prevent any damage/ injury to teats.
- Consistent milking at regular intervals using a fast but gentle technique, which empties the udder completely are important aspect.
- Always discard the milk from the animal treated with antibiotics for the prescribed days.
- Avoid noise during milking because it will have a negative influence on the release of milk.

F. Hygiene of milk utensils

Milk utensils should be designed, constructed, installed, maintained and used in a manner that will avoid the introduction of contaminants into the milk. Unclean milk utensils play a vital role in affecting the hygiene and quality of the milk. Therefore, the utensils and equipment used during milking should be:

- Preferably made up of non-rusting and non-absorbent materials (i.e. Stainless steel, aluminum or galvanized iron).
- Smooth, have minimal joints, free from dents and cracks.
- Milk handling equipment should be cleaned or washed immediately after milk removal.
- The detergents and chemicals used should be non-injurious, and non-abrasive in nature.
- Regularly maintain and sanitize milking equipment according to manufacturer and vet recommendations.
- Properly cleaned vessels should be placed in inverted positions.

6.2. Hygienic milk handling and transportation

6.2.1. Hygienic Milk Handling

Milk is very rich in nutrients because of this, the bacteria that cause spoilage can grow very quickly in milk. If milk is stored for a long time especially in high temperature, microorganisms like bacteria multiply very fast and the milk will have high number of bacteria which spoil quickly. Therefore, handling of milk strongly affects the hygiene and quality of milk and milk products. Points to follow to keep the number of bacteria in milk low and avoid milk spoilage:

- Handle milk in a cool and clean place after milking. The low temperatures reduce the rate of growth of the spoilage bacteria.
- If you do not have a refrigerator or cooler, store milk in a cold-water bath or wrap the milk can with a wet sack or blanket, but ensure the milk container is well covered to prevent dirt from entering the milk.
- Pour the milk into a container through a fine metal gauze strainer or muslin cloth.
- When transferring milk between containers, pour the milk instead of scooping. Scooping may introduce spoilage bacteria.
- Do not handle milk if you are sick especially with communicable disease.
- Don't keep milk for a long time before it is delivered to the collection point/center.
- Metal containers are preferable because they are and are durable, easy to clean and sanitize.
- Always use certified food-grade containers for milk handling, e.g., aluminum, stainless steel or food-grade plastic cans.
- Non-food grade plastic cans, buckets and jerry cans must not be used.
- Make sure that milk handling equipment meets the following conditions:
    - Be easy to clean and disinfect.
    - Have a Smooth finish without cracks or rust stains.
    - Have wide opening. This makes it easy to clean.
    - It is used only for milk, not for anything else.
    - Do not store milk in containers that previously contained paint, herbicides and other chemicals because traces of these substances can taint milk.



a) Aluminum milk cans



b) stainless steel milk cans



c) standard plastic milk can

Figure 26. Types of proper milk handling equipment

Milk storage area

As a food storage area, the milk storage room needs to be managed to minimize the risk of contamination.

- Milk storage room and areas must be kept clean at all times.
- The premises/buildings used for storing milk should be clean, pest-free, and well-ventilated with adequate lighting and protected from dust, rain and direct sunlight.
- Milk should not be stored in the same room with agricultural produce (e.g. Onions) or chemicals like paint or paraffin, which can taint the milk with off odors.
- Doors should be kept closed. Vermin, birds and other animals must be excluded.
- The structure of the milk storage room must not expose milk to the risk of contamination.
- The milk storage room should not be used for purposes other than the cooling and storage of milk, and the cleaning and storage of milking equipment

Hygienic milk collection

Milk collection is an activity of collecting milk from producers/ cooperatives/traders to a center where organized facility of bulk cooling and handling is available. Once the milk from several group members is collected in a central location, the milk can be processed or transported to processing centers or markets. Milk can be collected through the milk collection point and milk collection centers. During milk collection the following should be considered:

- Always keep the milk covered to avoid airborne contamination.
- The place of milk collection should be maintained clean.
- Cleaning agents should be kept away from the milk collection place.

- Lids of cans should not be kept on the ground and should be tightly fitted on cans to prevent spoilage and spillage of milk.
- Ensure that milk cans are kept in shade before loading in the truck.
- Trucks without covers or lids should not be used, as they will expose the milk to sunlight and hence, accelerate the growth of microbes.

Milk cooling

Milk chilling/cooling means holding milk below 4oC at production and collection centers to preserve the quality as produced until it is transported and distributed. The equipment used for initial chilling/ cooling of milk depends on variety of variable factors such as quantity of milk and availability of chilling facilities. In general, the milk chilling at a production farm or collection center can be done in two ways i.e., Can cool and Bulk milk cooling.

i. Can Cooling

- In this method milk is stored in cans after milking most of the time until it is used or processed.
- It can be by immersion of milk cans in chilled water insulated tank and Surface Coolers
- The primary energy source for milk can immersion in chilled water systems is electric power.



Figure 27. Immersion of milk cans in chilled water insulated (transitional)

ii. Bulk Milk Cooling

Electrical (generator) energy source type: When the production of milk is in thousands of liters, bulk milk coolers can be used. These are easy to clean and wash and remain hygienic in use. This tank is permanently fixed in an outer metallic tank insulated from the outside.



Figure 28. Bulk milk cooler

Solar energy source type (Solar thermal): Containerized, stand-alone, photovoltaic milk cooling tank system applied where electricity is not available, and it is an economic technology. A system transforms solar radiation into energy for heating, cooling or mechanical applications.

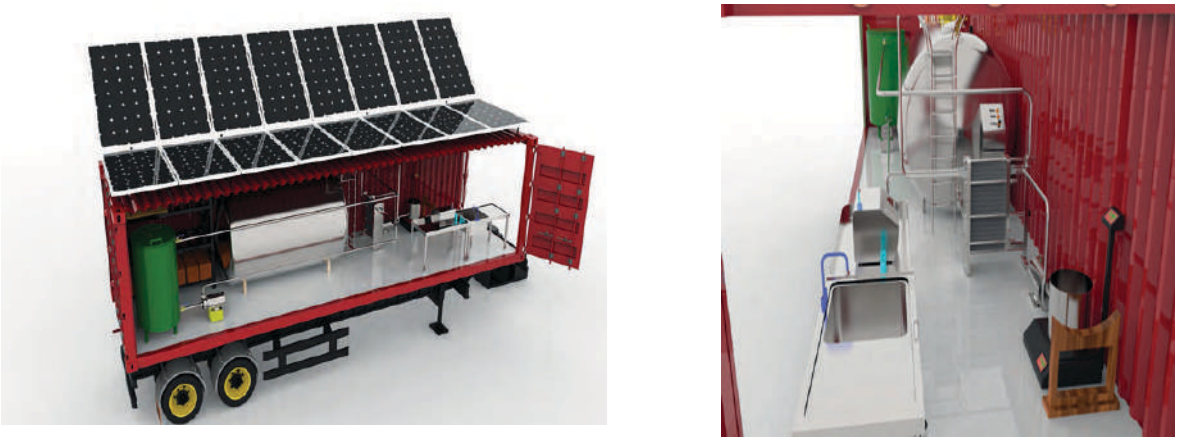


Figure 29. Containerized bulk milk cooling center using solar panels

6.2.2. Hygienic Milk Transportation

Milk transporters play an important role in transporting milk along the dairy value chain. They may transport milk in cans or bulk tankers. They carry large volumes of milk and therefore bear a huge risk if the milk gets spoilt. Milk transporters should be able to:

- Carry out basic quality tests of milk.
- Distinguish between poor and good-quality milk.

- Transport milk in appropriate containers and in a timely manner.
- Maintain the good quality of the milk during transportation.
- Operate a milk transport business in compliance with regulations and good hygienic practices.
- Maintain accurate records.

Milk transportation equipment: Milk can be transported from producers to processors and/or consumers. It is a very perishable product; transporters must ensure high levels of hygiene, speedy transport and careful handling.

- For small quantities of milk, the ideal milk transportation equipment would be metal milk containers made from aluminum and stainless-steel containers.
- The transport milk container should have a lockable lid to prevent spillage.
- Non-food grade plastic cans, buckets and jerry cans must not be used.
- Transportation of larger quantities of milk requires insulated bulk tankers. The tanks can be connected behind a truck.



Figure 30. Milk transportation equipment

6.2.3. Basic milk quality testing

Milk quality testing is performed to ensure the safety and hygiene of milk throughout the entire milk value chain, from the farm to the processing plant. Milk quality test is the use of various tests to ensure that milk is safe, healthy and meets the standards for chemical composition, purity and levels of bacteria and other microorganisms. There are various basic milk quality tests such as organoleptic test, Clot on boiling test, Alcohol test, lactometer test and Chemical Composition Test (Digital milk analyzing methods).

Organoleptic Test: The organoleptic test permits rapid segregation of poor-quality milk at the milk receiving point/centers.

Clot on Boiling (COB) Test: This test is quick and simple and allows you to reject milk that has developed high acidity pH less than 5.8 or Colostrum or mastitis milk that has very high percentage of whey proteins, which do not withstand heating at high temperatures in milk heat treatment.

Alcohol Test: The test is quick and simple. The alcohol test is more sensitive than the Clot on boiling test (COB). Milk, which pass during Clot on boiling test analysis, may fail by the alcohol test since it detects milk whose pH is about 6.4 while COB detects up to 5.8 pH Colostrum and mastitis milk may also detect positive to the alcohol test.

Specific gravity measurement (Lactometer test): Adulteration of milk may be defined as addition of any material to the milk, or removal of any constituent of milk. Adulteration of milk is not an allowed practice but milk producers/collectors intentionally added some adulterants to the milk to increase its volume to increase their income. The common adulterants in milk are water, removal of fat, addition of cane sugar, starch / cereal flour, skim milk powder, gelatin, etc.

Digital milk analyzing methods: There are many types of digital lab equipment's (machines) developed for further analyzing milk and milk products composition within short period of time. For example, lacto scan lacto scope, dairy food scan, eco milk and etc.

7. Dairy farm Record keeping

Record-keeping is a fundamental aspect of effective dairy farm management. It involves systematically collecting, storing, and managing information to enhance productivity, animal health, and overall farm profitability. Timely and reliable records are required for effective decision-making to follow efficient management for realistic production.

7.1. Importance of record keeping

Farm records are a necessary element of good dairy management. Records keeping are important for the following reasons: -

- Dairy farm records provide the basis for tracking & evaluating performance.
- Taken timely action with regard to cattle breeding, calving, disease, production and feeding.
- Provide up-to-date information for decision making, both at operational and strategic aspects of the farm, policy makers, research work and extension purposes.
- Helps in overall better supervision and management of the herd.
- Helps in determining the income and expenditure (economics) of a dairy farm.
- Helps in estimating the cost of milk production.
- Helps to compare the herd performances in different years
- Helps to identify problems/gaps and setting future goals/directions for the farm.
- Inform farmers about strengths and weaknesses in their farm operation.
- Improve traceability.

7.2. Criteria for good record keeping

If records are not to be more troublemakers than they are worth to farmers, they should satisfy the following criteria:

- Records must be useful: Unless data which is being recorded will at some future time be used (turned in to information) in making management decisions, it should not be recorded at all.
- Records must be kept as it can be easily converted into information: Before keeping a record, the eventual end use must be decided before the data are recorded for later analysis and interpretation.
- Record keeping systems must be simple: Dairy farmers have enough to do without

- burdening themselves with complex record keeping systems which are difficult to understand and time consuming to complete it.
- Duplication must be avoided as much as possible: Some data may have to be recorded more than once in different forms but this must be reduced to a minimum.
- Records must lead to actions being taken: Information must lead to quick actions.

7.3. Types of dairy farm recordings

There are different types of dairy farm recordings. The following list provides the most applicable types of dairy farm recordings at smallholder farmer level in the Ethiopian context:

a) **Animal Identification/history record -** provides information about major events occurred on individual dairy cattle from birth up to the time it leaves the herd. It includes Animal Name or ID, date of birth/purchase, breed type, calving period (month) for each calving interval, lactation period (days) for each lactation, lactation yield (average per day and total yield), date of drying, calf gender and ID number, age when culled, date of disposal (sales/death) see the record format in the anne3.



Figure 31. Animal Identification/history recording

b) AI service Record – help as a guidance to determine the success of AI service and profitability of the cow calf operation when accomplished with accuracy and consistency. It shall include owner's profile, Animal identification number/ name, Dates heat observed; Service Dates; Sires ID and Breed; Technician`s ID; Service number (First or Repeat); pregnancy examination, previous date of calving; expected calving date, drying off date, and any additional remarks.

c) Breeding Records: it should include ownership detail profiles; Name and/or Identification number of animals; Identification method (Ear Tag, tattoo etc.); Birth date or date purchased

and Breed Type. Pedigree Information such as extended family (Names & ID), Dam; Sire; Sire`s Sire; Sire`s dam; Dam`s Sire and Dam`s Dam, and Breeding parameters such as Age at First Calving, Parity, Date of Calving; Days after last calving, Calving complications; Calving Interval, reproductive diseases/disorders (if any) have to be recorded.

d) Production Record - Milk production record captures the individual cow milk yield per day, and has to include milking cow name or ID number, daily milk yield, and start & end lactation period/date.

e) Feed & Feeding Record – dictates about feed produced and/or acquired storage and treatments, and feeding system. Feeding records give information about the amount, type and quality of the feed provided to dairy animals, type and quantity of on farm feed produced and feed purchased from outside sources, and events happened following feed changes.

f) Health (Disease and treatments) Recording: Information management is fast becoming the key to effective action in animal care and keep track of the disease events in each animals' lifetime; guide to better management practices by leading the attention to repeated events or seasonality or certain vulnerable groups of animals over time (e.g., it can show how animals almost always need disease treatments in different seasons). Accordingly, health record should include disease occurrences, treatments and control measures practiced, events that affect welfare of the cows that interferes with reproductive performances and calves' survival.

g) Financial Recording: Financial recordings are records of the costs and earnings related to the dairy farming, recorded for financial analysis and dairy enterprise appraisal.

h) The necessary dairy farm record-keeping template and formats are listed below in Annex 3.

7.4. Culling Practiceon on Dairy Cattle Farms

Culling in dairy farming refers to the process of removing cows from the herd, whether for sale, slaughter, and typically involves replacing them with new animals. It's a routine practice in dairy operations, often driven by a combination of factors like low production, health issues, or reproductive problems, and is crucial for maintaining herd health and profitability.

Reasons for culling: -

- Low Production: - Cows with low milk yield, short lactation length, long dry period and sharp decline in yield are to be culled.
- Reproductive failure: - Cows with long anoestrous, long service period, repeat breeding, Cystic ovary, recurrent abortion, and higher age at first calving are to be culled.
- Physical defects: - Defects lead to low production and reproduction. e.g., loss of teat/teats. Mastitis, eversion of uterus/vagina, permanent lameness, loss of horns, old age, cryptorchid, pendulous udders, poorly balanced quarters, emaciation. Deviation from breed type, etc., chronic diarrhea.

- Bad behaviours - Animals with fighting habits, kicking at the time of milking & treatment, suckling of own milk, biting with horns, not allow the service, pica, soiling of feed material etc.
- Farm space problems: smart Culling Example: Low producers from the herd.

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Annex 1

Table 1: Reproductive and Production Effects of Macro and Micro Mineral Deficiencies in Dairy Cattle

Type	Mineral	Key Deficiency Effects
Macro-minerals	Phosphorus (P)	↓ Feed intake, ↓ milk yield, irregular estrous cycles, ovarian inactivity, cystic ovaries
	Calcium (Ca)	↓ Muscle contractility, ↓ dry matter intake, milk fever, retained placenta, poor fertility
Micro-minerals	Copper (Cu)	Delayed puberty, repeat breeding, low conception, embryonic loss, poor semen, ↓ and libido
	Selenium (Se)	etained placenta, abortion, stillbirth, irregular estrus, cystic ovaries, mastitis, metritis
	Manganese (Mn)	Silent estrus, cystic ovaries, delayed ovulation, ↑ embryonic mortality, ↓ conception
	Zinc (Zn)	Abortion, low birth weight, delayed puberty, prolonged labour, and implantation failure

Source: McDowell, 2023; Feyissa, 2023.

Table 2: The essential vitamins, their sources, deficiency, and their consequence in dairy cattle are described

Required vitamins	Source	Deficiency	Critical stage
Vitamin A	• Fresh grass (natural pasture, Napier, Desho, Rhode's grass) • Sweet potato vines • Cowpea and lablab forage • Green maize (early stage) • Legume forages (e.g., Alfalfa • Vetch, etc.) • Cassava leaves • Carrots • Commercial Premixes / Mineral-Vitamin Blocks	• Night blindness, • weak immunity, • retained placenta, • poor conception	• Late pregnancy, • early lactation, • calves
- Vitamin D	• Commercial Premixes / Mineral-Vitamin Blocks • Sunlight Exposure facilitate Vitamin D synthesis in the skin	• Rickets in calves, • milk fever, • weak bones	• Especially for growing calves (bone development), • Pregnant cow calcium regulation lactating cows,
Vitamin E	• Fresh grass (natural pasture, napier, desho, rhode's grass) • Sweet potato vines • Cowpea and lablab forage • Green maize (early stage) • Legume forages (e.g., alfalfa, • vetch, etc.) • Commercial premixes / mineral-vitamin blocks	• muscle degeneration, • reproductive failure	• Transition cows,lactating cows
Vitamin K	• Fresh grass (natural pasture, napier, desho, rhodes grass) • Sweet potato vines • Cowpea and lablab forage • Green maize (early stage) • Legume forages (e.g., alfalfa, • Vetch, etc.) • Commercial premixes / mineral-vitamin blocks	• Prolonged bleeding, • haemorrhage	• Calves, • stressed animals
Vitamin B	• Agro-industrial by-products: molasses, like wheat bran, brewer by-products, molasses, etc. • Commercial premixes / mineral-vitamin blocks	• Poor appetite, • low milk yield, • nervous disorders	• High-producing cows, • stressed or sick animals
Vitamin C Ascorbic acid)	• Synthesized in the rumen by microorganisms	• Rare deficiency, but helpful during heat or disease stress	• Calves, • heat-stressed animals

Source: McDowell, 2023; Feyissa, 2023.

Table 3: Recommended feedstuffs and feeding practices for different stages of lactating cow

Stage of lactation	Feedstuffs Recommended	Feeding Focus	Feeding Practice
Early lactation	• oat-vetch mixture, vetch hay, napier grass, alfalfa, elephant grass, tree lucerne, concentrates, maize and legume silage, treated crop residues, mineral supplements	• Maximize energy & protein intake	• Feed concentrates after milking, • Offer ad libitum access to clean water, • Monitor body condition and milk yield to adjust feeding levels • Avoid physical activity during peak lactation to prevent energy deficits
Mid lactation	• forages, hay, legume silage, and concentrate supplementation • Use crop residues supplemented with protein-rich by-products • maintain supplements	• Maintain production and body condition	• Balanced ration with gradual reduction in concentrate feeding •
Late lactation	• good-quality forages (native or improved hay). • lower lower-concentration supplement, minerals supplement should be offered	• Regain body condition	• Reduce concentrate feeding further • Prepare cows for dry period feeding management • Mineral supplementation remains important
Dry period	• Moderate quality forages (Vegetative stages of Napier, Rhodes, and desho grasses), mineral balance feeds	• Maintenance, prevent metabolic disorders	• Control calcium, avoid overfeeding

Table 4: Feeding schedule for intermediate weaning

Age	Whole milk Kg/day	Concentrate/calf starter Kg/day	Roughage Kg DM/day
1-4 days	Colostrum	Handful	
5-7 days	4	Handful	Grass or hay or Napier Leaves Ad.lib
2nd Weeks	4	0.5	
3rd Weeks	5	0.75	
4-5 Weeks	5	0.75	
6-9 Weeks	5	1	
10-11 Weeks	4	1	
12-13 Weeks	3	1	
14-15 Weeks	2	1	
Total	415	70	40 kg DM

Source: HARC has indicated that a calf fed with milk that has been partially replaced with skimmed milk was able to wean at the 70th day of its birth.

Table 5: Feeding schedule for early weaning

Age	Whole milk Kg/day	Concentrate/calf starter Kg/day	Roughage Kg DM/day
1-4 days	Colostrum		
5-7 days	4		Quality roughage green/hay or Napier Leaves Ad. Lib.
2nd Weeks	4-5	Handful	
3rd Weeks	4-5	Handful	
4-5 Weeks	4-5	0.5	
6-7Weeks	4-55	0.75	
8 Weeks	3-4	1	
9 Weeks	3	1	
10 Weeks	3	1	
Total		**55**	

Table 6. Early weaning feeding schedule using partially skimmed milk replaced whole milk

Source:-HARC has indicated that a calf fed with milk that has been partially replaced with a skimmed milk was able to wean at the 70th day of its birth.

Annex 2.

Subclinical mastitis CMT and test procedures

- Milk and discard the first three spades of milk from all four quarters on strip cup to visualize the flakes and clot in mastitis milk on black background (Fig 14a)
- Again, take milk from each quarter into different CMT rack quarters and mark with its respective quarters.
- Mix equal amount of CMT reagent with milk samples to each rack quarters (Fig 14b).
- Stir gently and observe gel formation.
- The more cells are in the milk the more "jelly like" reaction, indicating the presence of mastitis pathogens/infection

Calves age (days)	Daily offer (liters)		
	Colostrum	Whole milk	Skimmed milk
1-4	3.0	-	-
5-15	-	3.0	-
16-20	-	4.0	-
21-42	-	2.0	2.5
43-63	-	1.5	1.5
64-70	-	1.0	1.0
Total		124	



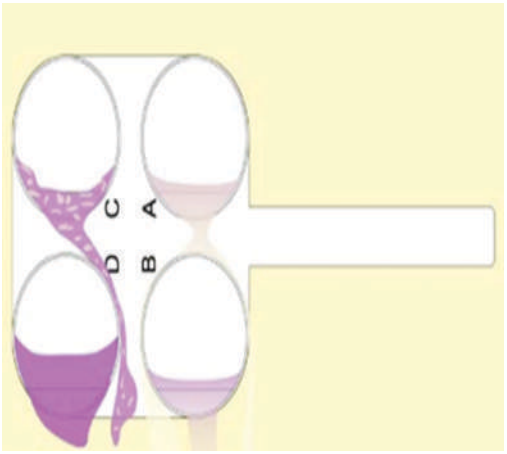


Figure 1. Screening test methods for subclinical mastitis

Annex 3.

Dairy Farm Record-Keeping Template

Table 1. Animal Identification/History Recording sample format

Name/ ID. No.	Date of Birth/ purchase	Breed Type	Calving period (Month)				Lactation period (days)				Calf get birth								Yield per Lactation period								Age when culled	Date of Disposal (sales/ death)
			1st Cl.	2nd Cl.	3rd Cl.	4th Cl.	1st Cl.	2nd Cl.	3rd Cl.	4th Cl.	1st cl		2nd cl.		3rd cl		4th cl		1st cl		2nd cl.		3rd cl		4th cl			
											Sex	ID. No.	Sex	ID. No.	Sex	ID. No.	Sex	ID. No.	Average Lt/day	Total yield lt	Average Lt/day	Total yield lt	Average Lt/day	Total yield lt	Average Lt/day	Total yield lt		

Table 4. AI data recording sheet

Individual Cow AI Record

Farmer: _____ Farm: _____

Address: _____ Cow ID: _____

Breed: _____ Breed of Sire: _____

Breed of Dam: _____ Milking only: Yes/No

Birth Date: _____ Lactation No: _____

Last calving date: _____ Suckling: for letdown only once Day/twice/day

Remarks: _____

Table 2. Breeding recording sample format

Cow Name/ID No.	Heat Date	1st Service		2nd Service		3rd Service		1st PD Test		2nd PD Test		Expected Date of calving	Dried off Date	Calving Date	Calf get birth		Remark
		Date	Method	Date	Method	Date	Method	Date	Method	Date	Method				Sex	Weight	

Table 3. Milk production recording sample format

Cow Name/ ID	Type of Breed	ctation Start Date	Milk yield per day in Litre																						ctation Start Date	Total yield
			Date		Date		Date		Date		Date		Date		Date		Date		Date		Date		Date			
			Morn. yield	Evn. Yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield	Morn. yield	Evn. yield		

Farmer: _____ Farm: _____
Address: _____ Cow ID: _____
Breed: _____ Breed of Sire: _____
Breed of Dam: _____ Milking only: Yes/No
Birth Date: _____ Lactation No: _____
Last calving date: _____ Suckling: for letdown only once Day/twice/day
Remarks: _____

Table 4. AI data recording sheet

AI No	Date	Heat to AI (hr)	AI time (am,pm)	Service of AI (U/C/V*)	Bull & Breed	Semen Batch	Date of Milk sample			Result of Milk Progesterone (mmol/l)			PD date & Result	Remark
							1	2	3	1	2	3		

(*U = Uterus; C = Cervix; V = Vagina)
Name of Inseminator: _____ Signature: _____ Date: _____
Laboratory Interpretation and Recommendation:
Name: _____ Signature: _____ Date: _____

Table 5. On-farm feed production/ purchase recording sample format

Type of feed	Year/Date of Production/ Purchase	Unit of Measurement	On Farm Feed Production		Feed purchased		Remark
			Quantity Produced	Area of land covered in Ha (for Forage & Natural Pasture)	Quantity purchased	Unit price	

Table 6. Feeding Recording sample format

Type of Feed	Unite of Measurement	Feed In take per day per cattle											
		Cattle 1		Cattle 2		Cattle 3		Cattle 4		Cattle 5		Cattle 6	
		Name/ID	Qty.	Name/ID	Qty.	Name/ID	Qty.	Name/ID	Qty.	Name/ID	Qty.	Name/ID	Qty.

Table 7. Animal health observation and treatment record

No.	ID/ Name	Date of observation	Major signs observed	Suspected disease	Treatment given			Response	Remarks
					Treatment	Duration	Dosage		

Table 8. Vaccination record

No.	ID/ Name	Date of vaccination	Type of vaccine used	Remarks

Table 9. Culling & Replacements

Date	Cow ID	Reason (Low Prod/Fertility/Health)	Sold Price	Replacement Heifer ID

